Children First, Housing Later*

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September 28, 2026

Abstract

Should housing assistance require sobriety? Without externalities, no: conditionality removes an option from forward-looking households. While conditionality reduces the externalities drug use imposes on users' children, it also raises homelessness, an externality that harms children and everyone nearby. A general-equilibrium model of homelessness, drug use, drug prices and mortality, calibrated to United States data, captures both externalities, with altruism matching how much less parents use drugs than non-parents. Full altruism sharply cuts drug use and homelessness. Conditioning assistance raises homelessness but substantially cuts drug use, raising welfare in transition and the long run through children's greater resources and future earnings.

Keywords: homelessness; housing vouchers; substance use disorder; intergenerational transmission; externalities; distortions.

JEL classification: I18; I38; H23; R21; J62.

1 Introduction

On a single night in January 2024 the point-in-time (PiT) count run by the Department of Housing and Urban Development (HUD) found 771,480 people homeless in the United States, the most since the count began. In July 2025 an executive order moved the federal government decisively to one side of a long argument about what to do about it: it required sobriety in federally funded housing and homelessness programmes.[1] The money followed, redirecting about a third of what the programmes at the time carried between them away from Housing First and towards programmes conditional on sobriety.[2] Whether housing assistance should be conditioned on sobriety is no longer an academic question. It is policy that decides the welfare of hundreds of thousands of people.

The argument between the Housing First and Treatment First camps is usually framed as a contest between two behavioural claims.[3] Critics of Housing First say that unconditional assistance is moral hazard: guaranteeing housing whatever a household does weakens the incentive to stay sober and attached to the labour market, so use and substance use disorder (SUD) rise, which feeds back to homelessness, because a SUD costs jobs and a lost job means not only more use but less ability to pay rent. Critics of Treatment First say that conditionality delays stable housing for exactly the households least able to wait, worsens their employment, and makes SUD recovery harder, because the street and shelters are poor places to recover. That too runs in a circle: worse recovery means more disorder, and more disorder means more homelessness. Both claims have merit, and the experimental literature has not settled between them, because the trials that could are short.

*I thank Ayşe İmrohoğlu, Kai Zhao, Joseph Steinberg, Efraim Benmelech, Jonathan Alschech, Ryan Quandt, Christopher Sharp and the seminar participants at the University of Toronto for their helpful comments. The data and replication code are available on GitHub at `https://github.com/srotberg/housing_first_children_later`.

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[1] Executive Order, *Ending Crime and Disorder on America's Streets*, 24 July 2025. Reported in `https://www.npr.org/2025/07/24/nx-s1-5479139/trump-homelessness-executive-order-civil-commitment-camping`.

[2] See `https://www.npr.org/2025/11/14/nx-s1-5553561/homelessness-housing-funding-trump-administration-hud`.

[3] For the case against Housing First as the federal government now makes it, see the executive order cited above; for the case for it, see the National Alliance to End Homelessness, `https://endhomelessness.org/blog/ceo-corner-week-of-july-28/`.

Randomised trials of Housing First find large and persistent gains in housing stability and mixed to null effects on substance use (Tsemberis et al., 2004; Aubry et al., 2016, among others), but they focus on the drug users themselves, and run six to twenty four months, so they cannot speak to long-run behaviour or to the budget required to fund either policy at scale. A model can follow what a trial cannot. It can trace the impact of a policy well beyond the households in the study group, to their children and to others in the economy, and well beyond the length of a trial.

This paper starts from the observation that the debate as framed is missing the mark, because the moral hazard both sides argue about has nothing to do with efficiency at all. Consider households that are forward-looking, as in Becker and Murphy (1988), and that have no self-control problems of the kind modelled with hyperbolic discounting at the moment they decide whether to start using. Suppose they account for the true hazard of acquiring a SUD when they make that decision, as the evidence suggests (Lundborg and Lindgren, 2002; van Ours, 1995; Grossman and Chaloupka, 1998; Saffer and Chaloupka, 1999; Caulkins, 2001; Gallet, 2014; Greenwood et al., 2025). Then at the same generosity and the same cost, Housing First weakly dominates Treatment First for every household. Conditionality removes an option from an optimiser and gives nothing back.

The case for Treatment First therefore has to rest on something outside the household's own objective, and this paper focuses on four externalities, which randomised trials cannot resolve: two that Housing First leaves in place and two that Treatment First creates.

The two Housing First leaves in place are harms drug use does to the user's children. The first runs through the household's budget, the *budget externality*: children live with their parents and share their consumption and their dwelling, but not the drug, so every unit a parent spends on drugs, and every job a SUD costs it, is consumption and housing the child goes without. When Treatment First reduces drug use, it diverts more resources towards children. The second runs through earnings in adulthood, the *earnings externality*: a child whose parent had a SUD during its upbringing starts adult life with lower productivity and earns less for the rest of its life. When Treatment First reduces the SUD, it raises the earnings of children in adulthood.

The two Treatment First creates both run through homelessness. By denying housing assistance to drug users, Treatment First exposes more of them to homelessness. The first externality this creates falls on everyone, the *aggregate homelessness externality*: every household's utility falls with the share of the population that is homeless, capturing in a reduced-form way the idea that it is less safe and pleasant to live in a place with more homelessness, and homeless households do not internalise the cost their own homelessness imposes on everyone else. The second externality falls on drug users' children, the *childhood homelessness externality*: a child lives wherever its parent lives, so a parent put in a shelter takes the child into the shelter too.

A parent takes the budget, earnings and childhood homelessness externalities into account only through its altruism, and only in part. How much weight it puts on its child is calibrated to the difference in drug use between parents and adults without children in the data. In the 2024 National Survey on Drug Use and Health (NSDUH), adults without children use drugs 18.3 percent more than parents with the same characteristics, so the weight is set so that removing a parent's concern for its child would raise drug use in the model by that much. The part of the child's loss a parent does internalise is thus pinned down by observed behaviour.

Given the degree of altruism, the size of each externality is set by data. The budget externality depends on how much a user household spends on the drug. The weight given to drugs in the utility function is set to match the share of their income that drug users spend on drugs, 16.2 percent, from retail drug spending (Midgette et al., 2019) and the incomes of adults who use in the 2024 NSDUH. The earnings penalty is set from a direct estimate: children exposed in utero to a Swedish alcohol experiment earn about 24 percent less as adults (Nilsson, 2017). The intergenerational transmission of SUD is then left untargeted: in the model a child of a parent with a SUD is 1.51 times as likely to acquire one itself, against 1.5 to 2.5 in the data (Mellentin et al., 2016; Yoon et al., 2013). The childhood homelessness externality carries no earnings penalty at all. The link between a parent's homelessness and the child's comes from intergenerational transmission alone, and the model is calibrated to estimates of earnings transmission from the literature (Solon, 1992; Mazumder, 2005; Chetty et al., 2014): a child's productivity is correlated with its parent's, and lower-productivity households are poorer and more likely to be homeless, so their children are more

likely to draw a lower productivity, be poor, and end up homeless as well. With this mechanism in place, a child of a homeless parent is 4.92 times as likely to be homeless itself, against 1.8 to 2.4 in the data (Bassuk et al., 1997). The aggregate homelessness externality is priced as a disamenity in flow utility, in proportion to the homelessness rate, set from the contingent valuation in Loubière et al. (2020), where the Europeans willing to pay anything to end homelessness would pay 0.20 percent of average annual earnings a year.

Who bears these externalities depends on who uses drugs, and one mechanism of the model decides it. Two facts about drug use have to be reproduced together, and the literature has so far explained them only by assuming that lower-education households, whose income is lower because of their education, have a stronger preference for drugs (Greenwood et al., 2025). Drug use is concentrated at the bottom of the income distribution, where the average user's income is about 74 percent of the average in the 2024 NSDUH, and it is not universal: most households do not use at all, so there is a margin at zero, with a participation rate of 8.90 percent among adults aged 26 and over in the same survey. To generate both I model drugs as a luxury good bought at a price, which puts most households at a corner where they optimally consume none. Left to its own devices, however, that same assumption concentrates use at the *top*, since the share of income spent on a luxury good rises with income. What reverses the income gradient, as I show, is the increased mortality risk that comes with a SUD: acquiring a disorder truncates the expected continuation value of life, and that loss rises faster than income does, so in equilibrium use concentrates at the bottom.

I proceed with two models, a stylized model for the intuition and a full quantitative model for the policy analysis. The stylized model is a life-cycle model with three periods of life (childhood, adulthood and old age) and rational forward-looking households, who care about consumption, housing and drugs. It shows four things at their clearest: how and why the income gradient in drug use reverses, how the value of life and moral hazard pull drug use in opposite directions, how one can find the optimal degree of homelessness, drug use and SUD, and why the externalities Treatment First removes outweigh the ones it creates.

In the model, an adult draws a productivity and an unemployment shock, and chooses how much of the drug to buy; drug users risk acquiring a SUD, and a SUD raises both the chance of dying before old age and the chance of losing employment. It chooses whether to live in a shelter or in a rental, and it may hold a means-tested voucher, which is rationed. Children live with their parents during the parents' adulthood and share their resources (except the drugs), and the weight a parent puts on its child is calibrated, as in the full model, to match the gap in drug use between parents and adults without children. The aggregate homelessness externality is in the model too, calibrated as in the full model. At the end of the parents' adulthood, their child becomes an adult, and its productivity draw is penalised if the parent acquired a SUD while the child lived with them. There is no government budget in the stylized model, so revoked vouchers are not offset by a higher probability of drawing a voucher for sober households; that is deliberate, so the mechanisms can be seen clearly, and the quantitative model takes up the detail. The model is calibrated to United States data.

Welfare is measured using the ex-ante expected utility of newborns, childhood included, in consumption-equivalent terms, against the world with rationed Housing First.

This stylized model shows starkly how the income gradient reverses. It separates the two penalties a disorder carries (death and unemployment). Scaled from nothing to 1.25 times each of their calibrated values, the death penalty takes the average user from _____ times average income to _____ times it; the unemployment penalty leaves it at _____. The reason is how each penalty varies with income. A fixed quantity of the drug costs a rich household less in utility terms than a poor household, which means that the utility price of using falls as income rises. A job loss cuts income by the same share for every household except those above average income, where unemployment insurance is capped, so it scales their utility from consumption by the same factor, and because households are risk averse, that costs a rich household fewer utils than a poor one. Thus, the unemployment penalty cannot pull drug use down the income distribution. In contrast, mortality destroys the whole continuation value, including the value of being alive at all, which is worth more to richer households, so when the death penalty is large enough it is the poor who use and the rich who do not.

The experiment is the reform itself. The programme stays at its benchmark rationing, with the same probability of drawing a voucher, and only the rule changes, from Housing First to Treatment First, which does not provide vouchers to households that use drugs or have a SUD. The stylized model keeps the price of drugs fixed; in the quantitative model it is endogenous.

Comparing each rule with no programme at all, and the two with each other, separates the two forces

housing assistance sets against each other. Under Treatment First there is no moral hazard, so only the value of life is at stake: assistance makes life worth more, and thus makes the risk of dying from a SUD costlier. Under Housing First moral hazard operates on top of the value of life, and thus the gap in drug use between the two isolates the moral hazard.

The first result of the stylized model is the optimum. To estimate the optimal degree of drug use and homelessness, I set the weight a parent puts on its child to one, so that nothing a parent does to its child is left external. In that economy nobody uses drugs, so nobody acquires a SUD, and homelessness drops by _____ percent.

The second result is that, without the externalities, the stylized model confirms the dominance result: _____ percent of households are strictly worse off under Treatment First and none is better off. However, counting what drug use costs the child, both the resources a parent diverts towards drugs and the earnings penalty of a parent's SUD, turns the ranking, from the first cohort of newborns on, even in the presence of the two homelessness externalities. This is because in this model the deterrence is large enough that no household holding a voucher chooses to use in the long run, and a household deterred by Treatment First from using never acquires the disorder, keeps more of its resources for its children, and its children inherit better draws, which more than outweighs the small increase in childhood homelessness, by _____ percent, from _____ to _____ percent of children (and the small increase in aggregate homelessness). In the transition, the first cohort born after the reform gains _____ percent of lifetime consumption, because its parents are deterred while it lives with them.

The full quantitative model that follows quantifies in more detail the mechanisms and the welfare implications of Housing First against Treatment First, and settles whether that ranking survives in a richer environment that fits the data. It is a general-equilibrium life-cycle economy, and households here, as in the stylized model, care about consumption, housing and drugs. Children live with their parents and share consumption and housing resources with them while the parents are 26 to 50 years old, and become adults after that. Households choose whether to live in a rental or a shelter, and whether to use drugs. Using carries the risk of a SUD, from which a household recovers only with a small probability each year: recovery is exogenous, harder in a shelter and with serious mental illness (SMI), but more likely in treatment, which a household can choose at a cost. SMI arrives exogenously with age and is more likely for households with a SUD, and like a SUD it lowers employment and survival, all consistent with the data (Milby et al., 2005; Olfson et al., 2015; Nielsen et al., 2017; Murrie et al., 2020; Hjorthøj et al., 2023).

The price of drugs clears the drug market, so drug prices respond to policy, and in general equilibrium, as drug prices decline, using drugs becomes more appealing, which is a force pulling against the deterrence effect of Treatment First. The government budget balances through the probability of receiving a voucher, so the programme's reach responds to what it costs. Conditionality is enforced by a drug test that the government pays for but that misses some users; a household that is caught loses its voucher and is barred from the programme for a time. The model is calibrated to United States data as well, with a benchmark of rationed Housing First, in which about a quarter of eligible households get a voucher with no sobriety conditions.

The model is also validated on a long list of moments that no parameter is fitted to, set out in Section 5.3. Among the larger ones: the homelessness rate by age band and the age composition of the homeless, both from the HUD PiT count; the share who return to homelessness within two years; drug poisonings as a share of deaths between 25 and 44; the earnings held by the bottom and top tenths and the income and net worth held at the very bottom of the distribution; the share of renters spending more than thirty and more than fifty percent of income on housing; the labour force participation of households with and without a disorder, and the income of those with one; how often the homeless enter treatment, and what they bring to it; the price elasticity of drug participation; and the odds that the child of a parent with a SUD acquires one, and that a homeless adult had a homeless caretaker. Each is a place the model can be caught.

The quantitative experiments start from full altruism, which shows how much is at stake: if parents weighed their children fully, drug use would fall by 36 percent and homelessness by 13 percent, even under rationed Housing First. The Treatment First reform itself is the same rationed change of rule as in the stylized model, with the difference that the probability of drawing a voucher rises to balance the budget. In the long run Treatment First cuts drug use by 30 percent and the SUD by 31 percent, but raises homelessness by 6 percent. The externalities it removes outweigh the ones it creates: a newborn gains 1.42 percent of

lifetime consumption, 50 percent of it during childhood, from its parent spending more on its consumption and housing instead of on drugs, and the rest as an adult, through higher earnings, even as childhood homelessness rises by 6 percent, from 1.02 to 1.08 percent of children. An earnings penalty for growing up in a shelter would not reverse this result: the share of children in shelters rises by only 0.06 percentage points, while the share of children living with a parent who has a SUD falls by 1.60 points, from 5.28 to 3.69 percent.

Accounting for the transition, the reform lands on an economy that already has many households with a SUD. Since recovery from a SUD happens only with a small probability, for many their SUD lasts a while, and their demand keeps the drug price elevated for many years. They are the first to lose their vouchers and be exposed to higher risk of homelessness, but the elevated price deters young adults who still have children living with them. In the first decade the number of drug users without a SUD falls by up to 48 percent and homelessness rises by up to 14 percent, against 29 and 6 percent in the long run, respectively. Children born at the reform gain about 2.1 percent of lifetime consumption, against 1.42 percent in the long run, because their parents are the young adults the elevated price keeps from starting, and their homelessness during childhood changes by _____ percent, from _____ to _____ percent.

2 Related literature

The literatures this paper draws on are set out in the order the argument needs them. The first three supply the machinery: what the trials establish about housing policy, how economists model the decision to use substances, and what structural models of homelessness already deliver. The next two supply the motivation, which is where this paper departs from its predecessors: what a parent with a SUD, or a parent in a shelter, transmits to their child, and how much weight a parent puts on the child who bears it. The last two are what the quantitative model is built out of: the hazards, taken one at a time from clinical and epidemiological work, and the drug market that makes the price endogenous.

Randomised trials of Housing First. A large empirical literature evaluates programmes that provide rapid housing without conditioning on sobriety, and finds large and persistent improvements in housing stability and reductions in shelter and other public service use (Tsemberis et al., 2004; Padgett et al., 2006; Larimer et al., 2009; Goering et al., 2014; Aubry et al., 2016). Evidence on substance use is mixed: most studies find no statistically significant change, and some report modest reductions in alcohol related outcomes. These evaluations typically measure outcomes over six to twenty four months and rarely beyond three years, which is short against a recovery process that Vaillant (1995) documents as gradual and protracted.

The model here offers a structural reading of that pattern rather than a dismissal of it: homelessness responds immediately to voucher design, while a SUD adjusts slowly, so a short trial is well powered for the first and not the second whatever the long-run effect is. The deeper limitation of a trial is not power but scope. A trial randomises households within a fixed environment, and holds fixed exactly the things that decide the welfare comparison: the price of the drug, the distortions of the taxes that pay for a programme at scale, and, most importantly, the outcomes of children who are not in the sample.

Economic models of drug use. Modelling use as forward-looking optimisation dates to Becker and Murphy (1988), and the assumption is empirically defensible at the margin where policy operates. Both users and non-users hold risk perceptions close to the objective probabilities (Lundborg and Lindgren, 2002), and Greenwood et al. (2025), who allow perceived and true SUD risk to differ, find the model implied beliefs are close to the true risk. The price evidence points the same way and is taken up with the drug market below, since the elasticity of participation is what disciplines the size of the behavioural response to Treatment First and to Housing First. A separate literature emphasises temptation and dynamic inconsistency (Laibson, 1997; Gruber and Kőszegi, 2001), under which present-biased households over-consume relative to their own long-run preferences and corrective policy has a rationale with no externality at all. That literature is complementary here rather than opposed, and it cuts in the same direction: a present-biased household would welcome a sobriety requirement as a commitment device, so introducing it would strengthen the case for conditionality rather than weaken it.

Where this paper departs from that literature is in what generates the income gradient in use. Greenwood

et al. (2025) build a macroeconomic model of the opioid crisis in which households differ by skill, and the concentration of use among the less skilled is carried in part by preference differences towards drugs across those groups. In contrast, in the model here no preference parameter varies with income: every household has the same weight on the drug, the same Stone-Geary endowment, the same hazard per unit used and the same mortality penalty. The utility function does not change with the state the household is in, or with the choices it makes. The gradient comes out of the mortality channel instead, because a SUD destroys a continuation value and the continuation value rises with permanent income, so the household with more to lose loses proportionally more than it gains from a drug that is cheap relative to its income.

Structural models of homelessness. The paper builds directly on İmrohoroğlu and Zhao (2022), who model homelessness as a rational response to income risk, mental health shocks, housing costs and housing policy. In that framework housing policy reaches homelessness through income loss and housing costs alone, so a sobriety-contingent policy has no behavioural margin to act on in their model. Adding endogenous use, disorder and recovery, together with state-dependent mortality, creates that margin: the same voucher that insures a household also changes what using costs it, and the disorder it may acquire then feeds back into employment, housing and survival. Abramson (2026) makes the complementary point on the other side of the market, showing that eviction policy changes landlord screening and rents and thus changes who becomes homeless in equilibrium. Here the causation runs both ways, and the two directions reinforce each other. On the one hand, the SUD costs employment, which costs a household its housing. On the other hand, losing housing worsens employment and raises mortality, which raises the value of using relative to the value of life being risked, since a household with less continuation left to lose risks less by using drugs. Each direction feeds the other, so what the model produces is a vicious circle rather than a one-directional flow.

Intergenerational transmissions. Three links have to hold for the argument to work, and each is estimated separately.

The first is that a parent's SUD changes the childhood. The administrative record of the child welfare system documents it at scale: parental substance abuse is recorded at more than a third of removals. The epidemiological literature documents it too. Household substance abuse is one of the adverse childhood experiences of Felitti et al. (1998), and children of a parent with a SUD are between one and a half and two and a half times more likely to acquire one themselves when one parent is affected (Mellentin et al., 2016; Yoon et al., 2013).

The second link is that the childhood shows up in the child's own adult earnings, and the mechanism is as follows. A parent who has lost their job to the SUD, and who spends what money is left on the drug, has less of it for the child. Time investment in the child goes the same way. What the child loses is schooling, a good peer group, and the ordinary attention a household gives, in a home more chaotic than it would otherwise be. The child reaches the labour market with less human capital and fewer good connections for it.

Each step of that is measured, and each has been built into a model before. Dahl and Lochner (2012) find that family income raises children's achievement, using the Earned Income Tax Credit to move the income, and Restuccia and Urrutia (2004) put roughly half of the intergenerational correlation in earnings down to parental investment in education, most of it early. Card (1999) puts the return to a year of schooling at around ten percent. And Nilsson (2017) closes the loop on a parental substance shock in particular: children exposed in utero to a Swedish experiment that sharply raised alcohol availability earn about twenty four percent less as adults and finish less schooling. On the transmission of the disorder itself, Mellentin et al. (2016) follow 398,881 offspring in the Danish registers and Yoon et al. (2013) a United States population-based sample, and both find a child with one affected parent likelier to acquire a SUD, at 1.44 and 2.51 respectively. What no study does is estimate the effect of a parent's SUD on a child's lifetime earnings directly.

The third is that housing, and the poverty that comes with bad housing, is one of the environments doing the transmitting, and it affects the transmission of earnings as well. Chyn (2018) finds that children displaced by Chicago's public housing demolitions earned sixteen percent more a year as adults. Chetty et al. (2016) estimate that a move to a lower-poverty neighbourhood before age thirteen raises adult earnings by thirty one percent, and the gain falls with the age at the move, which says that what is transmitted is

the environment a child is raised in rather than anything the child carries. Bassuk et al. (1997) measure the transmission of homelessness itself. Its case-control design is what makes it usable: among 220 homeless mothers and 216 low-income housed mothers, a homeless caretaker is likelier in the homeless arm on both the mother's and the father's side, at odds ratios of 1.80 and 2.41.

What this paper adds is not another estimate but a price: it puts these magnitudes inside an equilibrium model where the parent's use decision is endogenous, so the transmitted loss becomes a quantity that policy moves rather than a coefficient.

How much a parent weighs its child. The externality is whatever the parent does not internalise, so its size is decided by the weight a parent puts on its child, and that weight has a literature of its own. Altonji et al. (1997) test pure altruism directly, on consumption within parent-and-child households, and reject it: the transfers parents make to their children while alive do not respond to the distribution of income inside the family the way complete altruism requires. Parents are probably altruistic, but not completely, which is the case for treating ω_c as strictly between zero and one rather than assuming either end of it. Their experiment cannot be run here. It compares parents against their own adult children, and every household in this model has a child, born only when it dies, so there is no parent against non-parent comparison to make in the benchmark. What the model can do is another test, and for this paper it is a more relevant one. In NSDUH 2024 past-month use among parents is 0.845 of use among non-parents, reweighted to the non-parents' joint distribution of age, income, sex and marital status so that the gap is behaviour rather than composition. The child term of the household's problem is the only thing in the model that makes a household use less than it otherwise would, so ω_c is set so that switching that term off raises use by the same 18.3 percent. The externality is therefore not assumed to be large, and its size is calibrated to observed behaviour on exactly the margin this paper is about.

What the hazards are built from. The model's transitions are not free parameters dressed as structure; each is anchored on a clinical or epidemiological estimate, and the calibration section names them one at a time. Four groups do the work. On mortality, Hjorthøj et al. (2023) put the excess death rate for a SUD at six to ten times the baseline, Baggett et al. (2013) measure it for the homeless by age, and Olfson et al. (2015) do the same for serious mental illness (SMI); against those hazards the value of a life comes from the willingness-to-pay estimates surveyed in Viscusi and Aldy (2003).

On the mental illness chain, Nielsen et al. (2017) identify the effect of a SUD on the onset of SMI in Danish registers, Gutwinski et al. (2021) meta-analyse its prevalence among the homeless, and Bradizza et al. (2006) review relapse under dual diagnosis.

On recovery, Vaillant (1995) documents how slow unassisted remission is, and Milby et al. (2005) supply the causal estimate this paper leans on hardest: a randomised trial on cocaine-dependent homeless adults in which the housed arm was abstinent 6.60 weeks against 4.48 in the unhoused arm, which is where both the treatment effect on recovery and the penalty homelessness imposes on it come from.

On employment and the cost of housing loss, Desmond and Gershenson (2016) estimate what losing housing does to holding a job, and Loubière et al. (2020) price the disamenity of visible homelessness through a contingent valuation, which is what makes the aggregate headcount enter anybody's utility at all. Taking these from outside the model is the point: the parameters this paper argues over are the ones in Section 5, and a reader who disagrees with one of them is disagreeing with the study behind it.

The drug market. Making the drug price endogenous is what lets a policy that changes the number of users change what using costs everybody else. Two elasticities govern it and they do different jobs, so the literatures behind them are worth separating: the demand side is driven by how much the extensive and the intensive margins respond to a given price, and the supply side determines how much the price moves when demand changes.

The price elasticity of demand. The demand side has two margins and the model produces both. The first is the extensive margin (participation), which is whether a household uses at all. And the second is the intensive margin, how much a user buys once using. Neither is calibrated. Both are produced by the model and checked against the literature in Section 5.3.

Both margins have been measured, and between them the estimates bound how far either policy can move use. Participation first. Estimates of the participation elasticity run from about -0.3 to -0.9 (Saffer and Chaloupka, 1999; Grossman and Chaloupka, 1998), and the second of those puts the mechanism in units: a permanent ten percent fall in the price of cocaine raises the number of cocaine users by a little over eight percent. The intensive margin is the smaller of the two, between about -0.3 and -0.8. That same ten percent fall raises the frequency of use among those users by a little over three percent, and Olmstead et al. (2015) put the conditional elasticity for heroin at -0.80, on two research designs that share a sample of regular users but not an identification strategy. Gallet (2014) spans both margins at once, a meta-analysis of 462 price elasticity estimates for illicit drugs covering the decision to use and how much a user then consumes, and finds the response larger for cocaine and heroin than for cannabis.

A separate point follows from the findings of van Ours (1995), Saffer and Chaloupka (1999) and Caulkins (2001): participation responds to price, even for drugs as habit-forming as heroin and cocaine, and long-run responses are consistently larger than short-run ones. That ordering is the signature of forward-looking behaviour rather than myopia, which is why it belongs with Becker and Murphy (1988) above as well as here.

The elasticity of supply. This is the general equilibrium offset, and it runs against Treatment First in particular: curing households takes them out of the market, demand and the drug price fall, and a cheaper drug draws others in. I am not aware of a direct estimate for an illicit drug. Cannabis is the closest market I found to an illicit drug, as it remains a Schedule I substance federally, so even a licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. Mace et al. (2020) estimate a price elasticity of supply of 1.46 on Washington State administrative data, and that is the elasticity the model reproduces.

In a nutshell. The trials establish that housing stability responds to policy but cannot price what happens in the long run, to others not in the sample, or to the distortionary tax system that pays for it. The rational addiction literature models the use and SUD decisions but abstracts from housing, children, and mental illness. The structural homelessness literature delivers a housing market equilibrium but has no drug margin for conditionality to act on. The transmission literature measures what a childhood costs but takes the parent's behaviour as given. This paper's contribution is to put them in one framework and then to insist on a distinction none of them needs on its own: that the moral hazard which dominates the policy debate is not a reason for conditionality, and that whatever case exists has to be made on the externality, on the tax, or on the design, one at a time and with the three kept apart.

3 A stylized model

3.1 Three questions

In the data, drug use is concentrated among low-income households, most of the homeless do not carry a SUD, and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has six state variables and fifty six parameters, which is what the welfare and fiscal questions require, but it also makes it hard to see which pieces of the environment do the heavy lifting. This note strips the environment down to help see the main drivers in the model, but where the main questions of this paper are still well posed.

Households are perfectly forward-looking, here and in the quantitative model. They know the true hazard of acquiring a SUD, they are aware of what carrying one would cost them, and they make no mistakes. Nothing below rests on misperception, myopia, self control, or on a taste that changes once the household has used. The model abstracts from government and from how the programmes are funded; both enter the quantitative model of Section 4.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution, and it is not universal there either: most poor households do not use at all. Both facts have to hold together. I model drugs as a *luxury* good bought at a price, which is what delivers the second: not all households

choose to consume them. However, left to itself the Stone-Geary form in equation (1) below means the share of income spent on them rises with income, so the environment concentrates use at the *top*. Two risks push the other way, and each costs the household something: a SUD raises the chance of losing work, which costs earnings, and it raises the chance of dying, which costs the value of the life it has left. The question is which of the two reverses the concentration from the top to the bottom, and I show that it is the risk of death.

2. **How does housing assistance shape the drug use decision?** Housing assistance pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: call this the value-of-life force, and it lowers use. In contrast, if the voucher is unconditional, then it cushions the consequence of acquiring a SUD, because a household that acquires one is housed anyway: call this the moral hazard force, and it raises use. Both are first order, so theory does not sign the effect of unconditional housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First better than Housing First?** In this model, without externalities, and where the two programmes cost the same to run, never. Housing First always raises welfare when agents are forward-looking, as they are here. A SUD creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension does not raise welfare, because Treatment First only removes an option that a household can take under Housing First as well. However, if a SUD generates externalities, such as affecting the earning potential of the children of parents with a SUD, then Treatment First may be preferred, on the evidence above. The model's work here is to show why Housing First raises welfare whenever there are no externalities (and no distortions from the programmes or from the taxes that pay for them, both of which the full model takes up), and then show how the externality that overturns it works.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and components that obscure the mechanisms have been removed and the model refitted without them. What remains are the main drivers of the results. The welfare evaluations below are for illustration only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits the data.

3.2 The environment

3.2.1 Households and preferences

There are two periods and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty six to eighty five. Parameters with a natural annual counterpart are compounded to that length: the discount factor is β per period, the gross interest rate is $R = 1/\beta$ so that $\beta R = 1$, and $\bar{\phi}$ is survival over thirty years. Notation follows the full model in Section 4.

A household is characterised by its productivity y, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma_y^2, \sigma_y^2)$, so mean income is one. Households differ in y, in age $j \in \{1, 2\}$, in employment status $e \in \{0, 1\}$, in voucher receipt $\eta \in \{0, 1\}$ and in SUD status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility depends on non-durable consumption c, housing services h and drug consumption v,

$$u(c, h, v) \;=\; (1 - \gamma - \theta)\log c \;+\; \gamma \log h \;+\; \theta\big[\log(b + v) - \log b\big] \;+\; \aleph. \tag{1}$$

The three weights sum to one, where the weight on housing consumption is γ and the weight on drug consumption is θ. Drugs enter only in the first period, since that is the only period in which I allow v to be chosen, and housing is chosen each period from a three point grid $\{h_1, h_2, h_3\}$. The pair (θ, b) governs the drug term: the marginal utility of the vth unit is $\theta/(b + v)$, so θ mostly sets how much a user uses and b whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, which is what generates non-users at all, because under plain $\theta \log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish when consumption is low. Households save in one risk-free asset and cannot borrow.

3.2.2 Timing

Period 1. Every household is clean. Employment $e \in \{0,1\}$ is drawn with the unemployment probability u_0, and a household that is income eligible draws a voucher with probability π_V; these are the same two lotteries it faces again in period 2. It then chooses consumption, housing, drug use $v \geq 0$, which for simplicity is the only period in which it chooses whether to use, and saving, paying market rent p_r on whatever unit it picks. It starts with no assets.

Using drugs generates the risk of a SUD, increasing in v,

$$\pi(v) \;=\; 1 - e^{-\lambda_d v}. \tag{2}$$

Because π is increasing in v, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

Acquiring a SUD and the death penalty. The SUD $d' \in \{0,1\}$ is realised with probability $\pi(v)$ for households that used and with probability zero for the rest, and the household survives into period 2 with probability

$$\phi(d') \;=\; \bar{\phi}^{\,\kappa_{\phi d}^{\,d'}}, \tag{3}$$

so $\kappa_{\phi d}$ is the mortality *rate* ratio with the SUD. The mortality penalty applies here, and here only.

Period 2. This is the terminal period, after which the household dies with certainty. A household without the SUD is unemployed with probability u_0 and one with the SUD with the higher probability u_1; if unemployed it receives income support (Section 3.2.3). The voucher is drawn again, subject to the policy rule (Section 3.5). The household chooses consumption and housing, there is no drug use decision, and assets are run down with nothing bequeathed.

3.2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the SUD. Income in a period, given permanent productivity y and employment status e, is

$$y^g(y,e) \;=\; \begin{cases} y & \text{if } e = 1, \\ \psi_{ui} \cdot \min\{y, \bar{y}\} & \text{if } e = 0, \end{cases} \tag{4}$$

with ψ_{ui} the replacement rate and $\bar{y}$ a cap on the earnings base. There is no other income risk.

Each period the household chooses

$$h_k \in \{h_1,\ h_2,\ h_3\}, \qquad h_1 < h_2 < h_3, \tag{5}$$

homelessness (h_1), a small unit (h_2) and a standard unit (h_3). Homelessness is free and delivers less housing services than the market does.[4]

A voucher covers part of the rent, is means-tested and is rationed. It is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0,1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \;=\; \pi_V \cdot \mathbf{1}\{y' \leq \kappa_V\, y^{\text{med}}\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{6}$$

where $\kappa_V\, y^{\text{med}}$ is the threshold for eligibility, y^{med} is median income, and a household holding one pays the fraction κ_r of market rent, so the rent paid on option k is

$$r_k(\eta) \;=\; p_r\, h_k \big[\kappa_r \eta + (1 - \eta)\big] \cdot \mathbf{1}\{k \neq 1\}, \tag{7}$$

which is zero when the household is homeless. Vouchers operate in both periods, and π_V, the share of eligible households actually covered, is a policy dial of Section 3.5.

[4] Two rented sizes is the fewest that lets a household economise on housing rather than lose it. With one, homelessness comes out at 26.22 percent against a target of 0.399, at any h_1.

The two policies differ in exactly one respect, the last indicator in equation (6). **Housing First** (HF) covers every eligible household regardless of (d', v). **Treatment First** (TF) covers only households that satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. The condition is applied to the period in which the voucher is paid, so what it tests differs by period: every household is born clean, so $d' = 0$ holds for everyone in period 1 and only v can fail the condition there, while in period 2 there is no drug choice, $v = 0$ holds by construction, and only the disorder can fail it. Since policy coverage in the states that follow drug use is the only difference, the gap between them is the moral hazard and the whole of it.

The condition binds in period 1 as well, and the timing there is worth stating. Every household arrives clean, so the period 1 voucher is drawn before v is chosen and conditionality cannot screen anyone out on entry. The order within the period is therefore: the household chooses v, the policy rule is then applied to the voucher it drew, and it chooses housing knowing what rent it faces. So under Treatment First a household that uses (period 1), or carries the disorder (period 2), is subject to full market rent immediately, in the period concerned.

3.2.4 The household's problem

Let a be assets carried in. Period 2 is terminal, so the household consumes everything and chooses housing,

$$V_2(a', y, e', \eta', d') = \max_{\{k\}} \Big\{ (1 - \gamma - \theta) \log c + \gamma \log h_k + \aleph \Big\}, \qquad c = Ra' + y^g(y, e') - r_k(\eta'), \tag{8}$$

and write $V_2(a', y, d') = \mathbb{E}\big[V_2(a', y, e', \eta', d') \mid y, d'\big]$ for its expectation over the period 2 employment and voucher draws, the SUD having already been realised at the transition. In period 1 the household draws (e, η) and then chooses housing, drug use and saving,

$$V_1(y, e, \eta) = \max_{\{k,\, v \geq 0,\, a' \geq 0\}} \Big\{ (1-\gamma-\theta) \log c + \gamma \log h_k + \theta[\log(b+v) - \log b] + \aleph + \beta\, \mathbb{E}_{d'} \Big[\phi(d')\, V_2(a', y, d') \Big] \Big\}, \tag{9}$$

with $c = y^g(y, e) - r_k(\eta) - p_v v - a'$ and $d' = 1$ with the use dependent probability of (2). Survival sits inside the expectation over d' because it depends on which d' is realised. A household uses if the value at its best positive v exceeds the value at $v = 0$.

3.3 Calibration

Every target below is documented in full in the replication package's data appendix, `data/DATA_APPENDIX.md`, which names the dataset behind each one, its year and universe, and the computation that turns it into the number quoted here. What follows gives the moment and its source; the appendix gives the arithmetic.

3.3.1 External calibration

Table 1 lists the parameters set outside the model. Preferences and demographics are conventional: the discount factor over a thirty-year period is $\beta = 0.860$[5] and the gross interest rate is $R = 1/\beta$, and survival over a period is $\bar{\phi} = 0.91$, from the Social Security Administration (SSA) period life table for twenty six to fifty five. For the SUD, the mortality rate ratio is $\kappa_{\phi d} = 8$, the midpoint of the range in Hjorthøj et al. (2023), so survival with the disorder is $\bar{\phi}^{\,8}$ over the period and the implied proportional fall in survival is $1 - \bar{\phi}^{\,8-1} = 0.48$, so a clean household's chance of dying before period 2 is 9 percent and a household that acquires a SUD has a 53 percent chance of dying before period 2.[6] The two unemployment probabilities are measured directly. $u_0 = 0.0485$ is the headline civilian unemployment rate from the Bureau of Labor Statistics

[5] The annual discount factor here is higher than the one in the quantitative model, and this is deliberate. A period here is thirty years, so every consequence of using drugs arrives thirty years after the decision, whereas in the quantitative model consequences arrive annually and compound. A low annual discount factor would weigh that distant death much less and the mortality channel would be losing to impatience rather than to anything the model is about, so a higher factor corrects for the coarseness of the horizon rather than making a claim about preferences.

[6] The second figure is not $\kappa_{\phi d}$ and should not be read as one: $\kappa_{\phi d}$ is a ratio of mortality rates, so survival goes from $\bar{\phi} = 0.91$ to $\bar{\phi}^{\,\kappa_{\phi d}} = 0.47$ and the chance of dying is one minus that.

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
Preferences and demographics			
β	Discount factor over a thirty year period	0.995 annually, compounded	0.860
R	Interest rate over a thirty year period	Set to $1/\beta$	1.162
$\bar{\phi}$	Survival over a period, no SUD	SSA period life table, ages 26–55	0.91
The SUD			
$\kappa_{\phi d}$	Fall in survival with a SUD, over a thirty year period	Hjorthøj et al. (2023), age 26+	0.48
u_0	Unemployment probability, no SUD	BLS CPS via FRED, 2014 to 2024	0.0485
u_1	Unemployment probability, with a SUD	NSDUH 2024, age 26+	0.1625
Income support			
ψ_{ui}	UI replacement rate	DOL, statutory UI rates	0.50
$\bar{y}$	Cap on the earnings base	DOL, state benefit maxima	1.00
Housing, drugs and the voucher			
p_r	Rent per unit of housing services	Normalisation	1.00
p_v	Price of a unit of drugs	Normalisation	1.00
h_3	Standard rental unit	Set	0.40
κ_r	Tenant's share of the rent	HUD voucher rules	0.30
κ_V	Voucher means test	HUD income limits	0.50
π_V	Chance of drawing a voucher	HUD 2023, recipients over eligible	0.266

(BLS) Current Population Survey (CPS), series UNRATE as distributed by Federal Reserve Economic Data (FRED), averaged over 2014 to 2024. And $u_1 = 0.1625$ is from the National Survey on Drug Use and Health (NSDUH) 2024 for respondents aged 26 and over.

Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules. Only the standard rental unit is set, at $h_3 = 0.40$ of mean income; the small unit h_2 is calibrated, on the share of housed households living in one. The voucher parameters are institutional: covered tenants pay 30 percent of market rent, eligibility is capped at half of median income, $\kappa_V = 0.50$, and $\pi_V = 0.266$ is the ratio of recipients to the income-eligible population in the 2023 Picture of Subsidized Households (PSH), published by HUD.

3.3.2 Internal calibration

Eight parameters are set inside the model, in Table 2, each matched to the moment it moves most rather than fitted jointly. The dispersion of log permanent income σ_y is matched to how unequal earnings are at the bottom: in the 2022 Survey of Consumer Finances (SCF), households aged 26 to 65 with positive wage and salary income hold 17.2 percent of total earnings in the poorer half of the distribution, and the quantitative model of Section 4 is calibrated to the same moment. The weight on housing γ is matched to the 26.8 percent of total expenditure that shelter plus utilities take in the 2024 Consumer Expenditure Survey (CEX); the model counterpart is housing over expenditure net of drugs, since the survey's total has no drug spending in it. The flow value of being alive $\aleph$ is matched to a value of a statistical life of 80 times average annual income, about \$9 million against a mean household income of \$113,000, which is the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars. The pull of the first unit b is matched to a drug participation rate of 8.90 percent from NSDUH 2024 for respondents aged 26 and over with a disorder of any severity, housing services in a shelter h_1 to a homelessness rate of 0.399 percent from the 2023 HUD PiT count, and the SUD hazard λ_d to the 3.68 percent of adults aged 26 and over carrying a moderate or severe SUD in NSDUH 2024.

The small rental unit h_2 is matched to the 37.3 percent of housed households living in a unit with fewer than three bedrooms, from Table B25042 of the 2023 American Community Survey (ACS), which is the source the quantitative model's size distribution comes from as well. The weight on drugs θ is matched to the drug spending share, the 16.2 percent of income users spend on the drug, which divides retail expenditure on the four drugs, from Midgette et al. (2019) for 2016 carried to 2024 dollars, by the family income of the using population in NSDUH 2024.

Table 2: Internally calibrated parameters

Parameter	Description	Target moment	Data	Model	Value
σ_y	Dispersion of productivity	Bottom half's share of earnings, SCF 2022	17.2%	17.2%	0.990
γ	Weight on housing services	Housing over expenditure net of drugs, CEX 2024	26.8%	26.7%	0.325
$\aleph$	Flow value of being alive	Value of a statistical life, Viscusi and Aldy (2003)	80.0	80.0	2.863
b	Whether a household uses	Drug participation, NSDUH 2024	8.90%	8.90%	0.00456
θ	Weight on drugs in utility	Drug spending over income among users, RAND 2016	16.2%	15.7%	0.2350
h_1	Service flow of homelessness	Homelessness rate, HUD 2023	0.399%	0.410%	0.01455
h_2	Small rental unit	Share of the housed in the small unit, ACS 2023	37.3%	37.9%	0.069
λ_d	SUD hazard per unit used	Adult SUD rate, NSDUH 2024	3.68%	3.71%	106.8

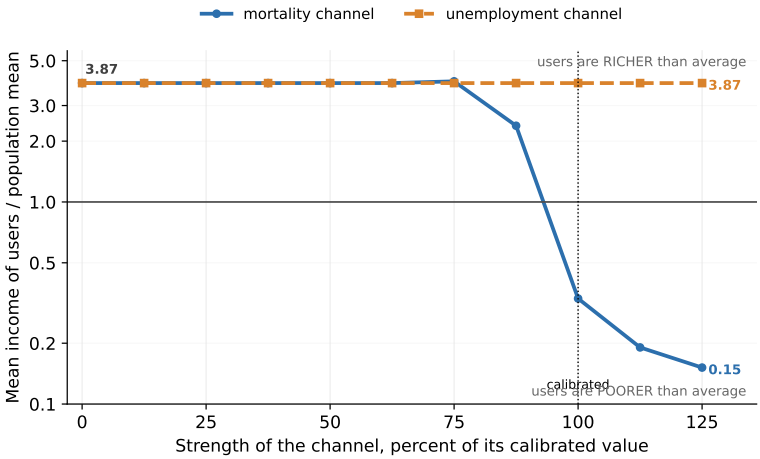


Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over the population average, on a log scale, and b is recalibrated at every point so that the model targets a participation rate of 8.90 percent at every point. The black dotted line marks the calibrated strength; everything to the right of it is extrapolation.

3.4 Who uses drugs

This section asks which feature of the model puts drug use at the bottom of the income distribution, and it starts from the force that any answer has to overcome. Drugs are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a SUD that is the only force operating, and drug use concentrates at the top: the average user then earns 3.87 times the population average.

Two features of the environment are candidates to reverse it. The *mortality channel* is that the SUD increases the chance of dying, through $\kappa_{\phi d}$ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose your home. Figure 1 scales each from nothing to 125 percent of its calibrated value, recalibrating b at every point so participation stays at 8.90 percent.

The result is sharper than the statement that mortality matters more. That channel takes the income ratio from 3.87 with nothing switched on down to 0.33 at full strength. At half of its calibrated value the ratio is unchanged at 3.87, and the reversal begins to happen over the last quarter of the channel. That threshold is a feature of the mechanism rather than an artefact. What mortality destroys is a continuation value, and a household only stops using when the expected loss of that value exceeds what the drug is worth to it; below the point where it does, a stronger penalty changes nothing about who uses, and above it the ordering starts to flip.

In contrast, the unemployment channel does not reverse it at all, leaving users at 3.87 times average income even fully switched on, and at 125 percent it is 3.87, still above one and no closer to turning. A

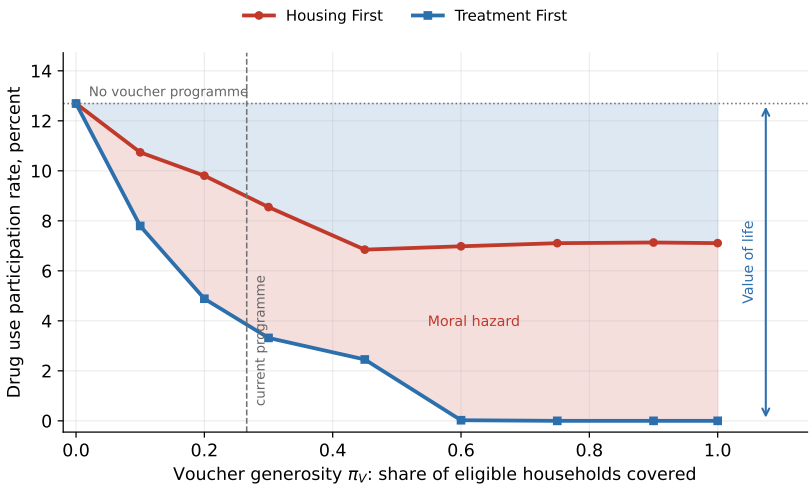


Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard force and the blue band above it is what Housing First delivers net; together they are the value-of-life force, marked at the right. The dashed line is the generosity of the current programme, $\pi_V = 0.266$.

resource penalty does not produce the concentration of drug use at the bottom.

Why should the two channels differ at all, when in both the disorder punishes the household? Because only one of them has a bite that grows with what the household has. Flow utility is logarithmic, so a *proportional* loss of income costs a number of utils that hardly moves with income: a household earning below the cap $\bar{y}$ that loses its job and falls back on income support gives up $-\log \psi_{ui}$, whatever it was earning. Multiplied by the $u_1 - u_0$ chance of it happening, the resource channel is close to a flat charge for using. What it has to overturn is not flat at all. Buying v units costs a rich household almost nothing in consumption terms and a poor household a great deal, so the utility price of using *falls* as income rises, and that is what puts use at the top to begin with. A charge that does not vary with income cannot reverse a price that does.

Mortality is a different object. What it destroys is the whole continuation value: the consumption of a life still to be lived, and ℵ, the value of being alive at all, which does not shrink as consumption does. That value rises with permanent income, so the expected cost of using rises with income at the same time as its utility price falls. Past a point the two cross, and it is the rich who stop first. The reversal is therefore not that mortality is the larger penalty; it is that mortality is the only one of the two whose size rises quickly with income. The household with more to lose loses more, and loses disproportionally more than it gains from drugs. This is the work of ℵ in equation (1), and it is worth saying why a constant can do it. What the household weighs against the drug is the cost of dying measured in *consumption*, which is the continuation value divided by the marginal utility of consumption. Flow utility is logarithmic, so that marginal utility is $1/c$ and a term worth ℵ utils is worth $c\aleph$ goods: constant in utility is *proportional to consumption* in the units the household actually trades off. The consumption part of the continuation contributes about $c \log c$, which is small and negative for a poor household, so it is the constant that carries the income gradient in the value of a life.

3.5 How does housing assistance shape the drug use decision?

The two forces of question 2, moral hazard and the value of life, can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value-of-life force operates. Under Housing First it is not, so both channels operate. Comparing the two policies at the same generosity therefore isolates the moral hazard force, and comparing Treatment First with no programme at all isolates the value-of-life force.

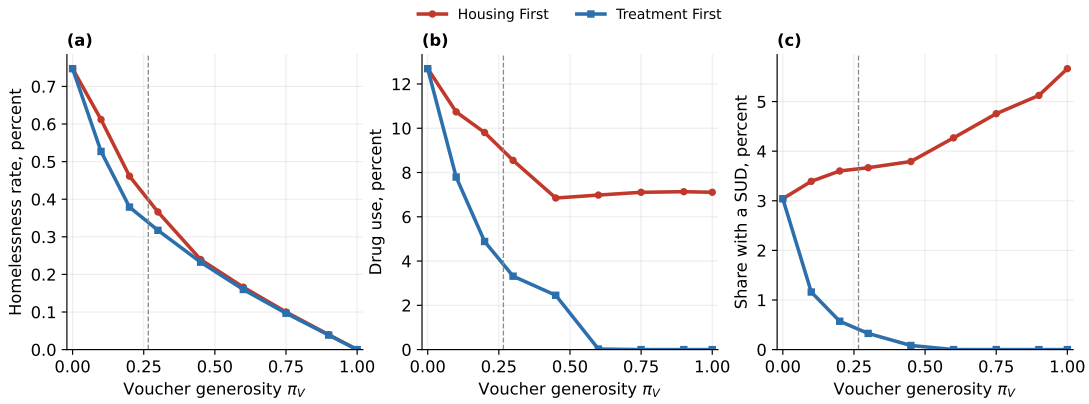


Figure 3: Homelessness, drug use and the SUD under the two policies

Figure 2 traces both as generosity rises, that is across values of π_V. Both comparisons here are between long-run steady states, each the rest point of the economy under a policy that has always been in place. Section 3.7 follows the path between them. With no programme at all, 12.69 percent of households use. Universal Treatment First takes that to 0.00 percent, a value-of-life force of -12.69 percentage points; universal Housing First leaves it at 7.11 percent, so the moral hazard force is +7.11 points and the net effect of Housing First is -5.58. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

Figure 3 plots the homelessness and SUD outcomes for each programme as its generosity changes, alongside drug use. In Panel (a) the two policies end in the same place on homelessness, both driving it from 0.75 percent with no programme to zero, and along the way Treatment First is if anything slightly below Housing First, because the households it deters from using never acquire the SUD that puts their housing at risk. Panel (b) repeats Figure 2 here for convenience. The larger difference is in Panel (c), where at universal cover Housing First leaves 5.663 percent of the population carrying the SUD and Treatment First leaves 0.000 percent. What the moral hazard buys is extra SUD cases among households that are housed either way.

Panels (b) and (c) move in opposite directions under Housing First, and the reason is that the programme acts on the extensive and intensive margins in opposite directions: it moves fewer households onto the drug, and those it does not deter it makes richer, so each of them uses more. Set against no programme at all, universal Housing First takes participation *down*, from 12.69 percent to 7.11 percent, while the share carrying the SUD goes *up*, from 3.041 percent to 5.663 percent. Fewer households use, but each one that does uses a great deal more: a user keeps its voucher whatever it consumes, so it is richer than the same household would be under conditionality, and the drug is a luxury. Since the hazard is $\pi(v) = 1 - e^{-\lambda_d v}$, what turns a user into a case is only the dose rather than the decision to use, and the share of users who acquire the disorder rises from 24.0 percent to 79.7 percent.

3.6 Welfare

3.6.1 Without externalities, Housing First always wins

Sections 3.4 and 3.5 established what each policy does to behaviour. This section asks what each does to welfare, comparing the two at the same generosity π_V, whose outcomes are in Panel A of Table 3. Welfare is measured throughout as a consumption-equivalent variation (CEV), set out formally in Section 6. The policies differ only in the last indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a household that optimises cannot be worse off under Housing First:

$$V_1(y;\ \mathrm{HF},\ \pi_V)\ \geq\ V_1(y;\ \mathrm{TF},\ \pi_V) \qquad \text{for every } y, \tag{10}$$

Table 3: Welfare at universal cover, with no externality

	Housing First	Treatment First
Panel A. Outcomes at universal cover		
Drug participation rate	7.11%	0.00%
Share with a SUD	5.663%	0.000%
Homelessness rate	0.000%	0.000%
Voucher outlay per household	0.0941	0.0937
Panel B. Treatment First against Housing First, no externality		
Households strictly worse off		7.4%
Households strictly better off		0.0%
CEV, those affected		0.70%
CEV, whole population		0.051%

Note: the CEV is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is expected spending on a household over both periods, the second discounted and weighted by survival, per unit of average income.

with equality exactly for households that never use under Housing First. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward-looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.

It follows that the moral hazard is not an inefficiency. Households here face the true SUD hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household optimally taking a risk it is now better able to bear. The +7.11 points of Section 3.5 are a statement about drug use and not about welfare.

Panel B of Table 3 puts welfare numbers on it: 7.4 percent of households are strictly worse off under Treatment First, none is better off, and those affected lose 0.70 percent of consumption. Those are exactly the households that use: a household that never uses faces the same voucher rule under both policies and is indifferent between them. The figure is not the 8.90 percent participation rate of Section 3.3 because the programmes compared here are universal, rather than the rationed benchmark that rate is measured at.

A fixed budget does not rescue the welfare case for conditionality either. Treatment First is only marginally *less* expensive than Housing First at universal cover, 0.0937 against 0.0941, so matching on cost rather than on generosity changes nothing. The reason both produce similar fiscal costs is the following. Survival into period 2 falls from 0.91 to 0.47 with a SUD, so the extra households Housing First puts into the SUD state die sooner and fewer are alive in period 2 to collect a voucher; against that, Treatment First withdraws the voucher from anyone who used, so it pays for fewer tenants while they live. The second effect is the larger, so the conditional programme ends up slightly cheaper, and the gap is small enough that nothing turns on it. So a case for conditionality in this model has to come from elsewhere.

3.6.2 An externality on children reverses it

One of the externalities modelled in this paper is the one imposed on the children of households with a SUD, and it runs through income rather than through preferences. I have considered two alternatives, but both seem to be at odds with reality. Letting the child of a parent with a SUD inherit a stronger taste, a lower b, matches a higher odds ratio for SUD for the child, observed in the data, but has counterfactually the wrong welfare sign: b enters only through $\theta \log(1 + v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. So the specification says that a child who inherits a stronger taste for drugs is *made better off* by inheriting it, and that is the opposite of what these children report. Growing up with household substance abuse is one of the adverse childhood experiences in Felitti et al. (1998), and it predicts higher rates of depression, worse self-reported health and lower life satisfaction in adulthood; the children of parents with a SUD also have lower educational attainment, which is the premise of the transmission literature δ_d, presented shortly below, is checked against (Mellentin et al., 2016; Yoon et al., 2013). A model in which the transmitted trait raises the child's own welfare cannot be describing any of that, whatever it

Table 4: The externality on children

Panel A. Transmission: what each income penalty implies			
δ_d	Child's SUD rate	Ratio	CEV
0.00	3.81%	1.00	0.0%
0.20	5.94%	1.56	41.2%
0.24 (baseline)	6.50%	1.71	52.7%
0.60	15.32%	4.02	281.1%
Panel B. Does the externality reverse the ranking?			
Private cost of conditionality			0.051%
Disorder cases prevented			5.663 pp
CEV to the child, one case			50.9%
Externality gain from conditionality			2.880%
Gain over loss			56.2

Note: Panel A is at the baseline programme, $\pi_V = 0.266$, since the transmission ratio in the data comes from an economy with a rationed programme; Panel B is at universal cover, one child per household, no discounting across generations. Every CEV here is the proportional rise in consumption, at every date and in every state, worth as much as the difference in lifetime value, so it is a compensating variation and can exceed one hundred percent.

does to the transmission rate.

Alternatively, letting it inherit a greater susceptibility, a higher λ_d, fails another way. A child facing a higher chance of converting use into a SUD optimally uses *less* ex ante, which is at odds with the data: children of parents with a SUD are *more* likely to use, not less.

In contrast to the above, income works on both counts, and it works through the channels explored in Section 3.4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use, which is aligned with the data. It is also supported by the empirical evidence, and it stands in for more than income alone: a parent who diverts spending and time to drugs invests less in the child's education and development. The income gradient does the whole job, so the model transmits a SUD across generations *without transmitting any preference for drugs*.

A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is a penalty

$$y_{\text{child}} \;=\; y_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent had a SUD when it died}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless when it died}\}}, \tag{11}$$

which the parent does not internalise at all. That is what makes it an externality.[7]

Both penalties are set from earnings estimates. $\delta_d = 0.24$ is Nilsson (2017), who follows children exposed in utero to Sweden's regional experiment in strong beer and finds they earn 24 percent less as adults. $\delta_h = 0.31$ is Chetty et al. (2016)'s 31 percent, the adult earnings gain from a move to a lower-poverty neighbourhood before age thirteen.

The two penalties also do different work: both policies drive homelessness to zero at universal coverage, so δ_h prices the programme against no programme while δ_d decides Treatment First against Housing First.

Panel B of Table 4 puts the two sides together and they are not close. The private cost of conditionality, in the first row, is 0.051 percent of consumption, small because only 7.4 percent of households are affected. Against that, conditionality prevents 5.663 percentage points of SUD cases in the second row, and preventing one of them is worth 50.9 percent of consumption to one child, shown in the third row. The gain in the fourth row is the product of those two, since one prevented case spares one child: 5.663 percentage points times 50.9 percent gives 2.880 percent of consumption, which is 56.2 times the private loss of the first row. That ratio exceeds one for every δ_d in Panel A except $\delta_d = 0$, where by construction there is no externality to price; at the smallest penalty shown it is still 44 times the private loss.

All of this is a crude illustration, so the impacts of policies can be unreasonably large. Its point is to show how an externality can reverse a welfare ranking that is otherwise airtight. Which way the comparison

[7] In the quantitative model the altruism of parents towards their children is calibrated rather than set to zero, so that the channel where a parent does take some account of what it does to its child is present and only the remainder is external.

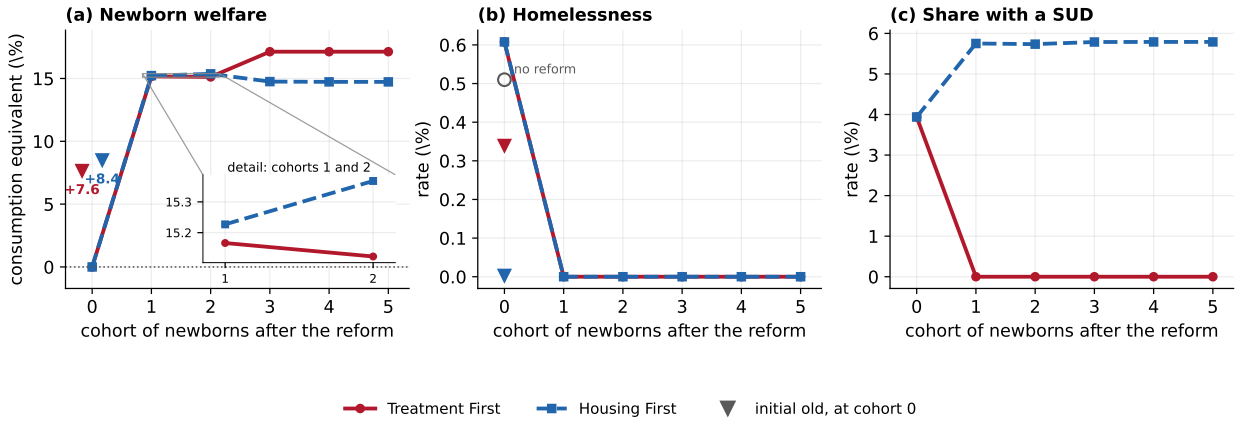


Figure 4: The transition after the reform

Note: Panel (a) is the expected lifetime value of each newborn cohort, as a CEV against the pre-reform rest point; Panels (b) and (c) are that cohort's own homelessness and disorder rates, so the welfare path can be read against the mechanism producing it. The reform moves the programme from Housing First at $\pi_V = 0.266$ to universal coverage under the policy named. Cohort 1 is born into the old income distribution and lives under the new rule. The marker in Panel (a) is the initial old, the generation already in period 2 when the reform lands, which no cohort line can carry: it is measured over its remaining period rather than a full life. Cohort 1 is the initial newborns, cohort 2 the second cohort of newborns, and so on.

actually goes is a question about magnitudes, and that is what the quantitative model of the next section is for.

The quantitative model carries two further channels absent here, and they are different in kind as well as in sign, which is another substantive reason the welfare ranking is a question of magnitudes. The first is not an externality at all but the distortions the programme creates and the distortionary taxes required to pay for it create; the second is an externality of the same type as the one on children, but not directly on children.

- **Programmes distort the decisions of recipients and are paid for out of distortionary taxes.** A programme lowers the labour supply of the households it does reach, because eligibility is means-tested, and the voucher is clawed back at a rate of κ_r, so a recipient who works more loses eligibility or some of the transfer. This, combined with the pure financing of the programmes, means that other households have to pay more taxes, which distorts their labour supply.
- **Homelessness is a disamenity in itself.** Households dislike homelessness around them and not only their own, so the aggregate rate is an argument of everyone's utility. This externality runs in favour of whichever policy houses more, which is the opposite direction from the tax.

3.7 The transition

Comparing rest points hides a key tradeoff. Treatment First takes immediately and gives back only later: at implementation a household with a SUD can do nothing about its condition but loses its voucher, and lower substance use pays off only for the generations after it. The link across generations is the penalty, not persistence of income: a child's income is a fresh draw times $(1 - \delta_d)$ if its parent acquired the disorder and $(1 - \delta_h)$ if its parent was unhoused when it died, so a cohort inherits a mixture over four types, no penalty, one of the two, or both, and that mixture is a Markov chain in the policy.

Figure 4 moves the programme from Housing First at $\pi_V = 0.266$ to universal coverage under each policy. Two groups are alive in the period it is implemented: the *initial old*, in their second period, who chose under the old rule and cannot recover in a terminal one; and the *initial newborns*, born to parents who finished before Treatment First was implemented. Welfare is a CEV against the pre-reform rest point, so "percent" means percent of lifetime consumption.

Each of the generations affected has a known sign here, with only its magnitude open. The initial

newborns can only be weakly worse off, since conditionality removes an option and gives nothing back and they inherit identically under both policies: they gain 15.17 percent against 15.23, so conditionality costs them 0.061 percent (see cohort 1 in Panel (a) of Figure 4).

The second cohort of newborns must be strictly worse off, because its parents are the initial old, whom Treatment First stripped of the voucher for a disorder acquired before it was implemented: 0.336 percent of it is born to a parent who was unhoused under universal Treatment First against 0.000 percent under universal Housing First, and the welfare gap widens from 0.061 to 0.245 percent (see cohort 2 in Panel (a) of Figure 4). What that cohort inherits identically is the *disorder*, settled before the condition existed and beyond any policy's reach after the fact, so only the housing margin differs.

The initial old are the ones conditionality costs most, and the figure is careful about what that means. The triangles in Panel (a) of Figure 4 price them against the same pre-reform world every cohort line is drawn against, and on that axis they gain under *both* policies, +7.6 percent under Treatment First and +8.4 under Housing First: universal coverage is worth a great deal even to a generation whose condition is an unchanged state. The gain is that large because the voucher is a rent subsidy to everyone it covers rather than a rescue for the few it takes out of shelters. A covered household pays 30 percent of market rent, and π_V is the share of the *income-eligible* who receive one, eligibility being capped at κ_V times median income. Raising π_V to one takes the share of this generation actually holding a voucher from 6.4 percent to 23.9 percent under Housing First and 21.9 percent under Treatment First. The two differ because an eligible household that carries the disorder fails the condition and holds no voucher however generous the programme is, which is conditionality's whole mechanism seen on one generation. Those covered are the poorest households in the model, where rent is most of what income has to cover, so their marginal utility of consumption is high and the subsidy's effect on their welfare, and thus on aggregate welfare, is correspondingly large.

What conditionality does is take 0.8 percent of that away from universal Housing First, 14 times what it takes from the initial newborns. This is because it puts 0.34 percent of them in shelters through the revocation itself, against 0.00 percent under universal Housing First and 0.51 percent had the old programme simply continued, which Panel (b) now shows alongside the cohorts. They are worse off than universal Housing First would have left them, and better off than before the reform, because universal coverage houses more of them than the rationed programme did, even with conditionality. That is the same figure the second cohort of newborns inherits, because the initial old are its parents: what conditionality does to them is not a cost that ends with them.

From the third cohort of newborns onward universal Treatment First raises welfare above universal Housing First, through the very cohorts it hurt: nobody uses, so nobody carries the disorder, and their children inherit better draws. Homelessness is zero for every cohort under both policies, since the voucher is means tested and every household poor enough to be at risk of ending up in a shelter is income eligible, so at universal coverage it holds one and can afford the small unit, so all the separation is in Panel (c), where the disorder share rises under universal Housing First and so more children inherit worse draws than under universal Treatment First, which has none.

To sum up, the cost falls on the initial old and the two cohorts behind them, and the benefit is reaped by every cohort after. Both policies lift the initial newborns by very nearly the same amount because the reform moves two dials of very different size: coverage is worth about 15.20 percent to them, conditionality costs only 0.06 percent, a ratio of 251 to one.

That a transition here *must* open with losses is a property of this environment. But the quantitative model is not bound that way, for two reasons absent here: its budget clears every period, so Treatment First, covering fewer households, taxes less the moment the reform lands and the early cohorts keep more of their income; and altruism is calibrated to a number greater than zero, so a parent internalises part of what its abstinence and the lower taxes Treatment First hand its lineage later on.

4 The quantitative model

This section describes the model the paper's numbers come from. Where the stylized model already carries something unchanged it is named rather than motivated a second time: the Stone-Geary form of the drug term, the logic that separates the two policies, and the transmission externality all appear below. Notation

follows Section 3 wherever the two models share an object: γ and θ are the housing and drug weights in both, b is the same Stone-Geary endowment, $\aleph$ the same flow value of being alive, λ_d the same SUD hazard, ψ_{ui} and $\bar{y}$ the same replacement rate and earnings base cap, r the same risk-free rate, and β is the discount factor.

4.1 Demographics and the state and choice sets

Households enter as adults, live at most J annual periods and retire at J_R, and j indexes age within that span. Survival from j to $j+1$ is $\phi(j, h', d, m, \tau)$ in (24), where it depends on age, the housing *chosen* this period, the SUD and SMI, and whether the household is in treatment, since treatment means it is not using, and thus is less likely to die. Newborns replace the dead exactly, so the mass of households is constant. The individual state is

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathcal{A} \times \mathcal{Y} \times \{1,2,3\} \times \{0,1\}^3, \tag{12}$$

where a denotes assets, y permanent productivity, e employment status, d the SUD, m SMI and η receipt of a voucher.

Each period the household chooses

$$\big(n,\, h',\, v,\, \tau,\, a'\big) \in [0,1] \times \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A}, \tag{13}$$

where n denotes hours, h' the housing unit chosen, of which h_1 is homelessness, v drug use, τ SUD treatment take-up, available only when $d = 1$ and requiring $v = 0$ in the period it is taken, and a' next period assets.

4.2 Preferences

Period utility is

$$u\big(c, h', v, n, \tau; d, m\big) \;=\; \frac{\big[\, c^{1-\gamma-\theta}\,(h')^{\gamma}\,(v+b)^{\theta}\,\big]^{1-\sigma}}{1-\sigma} \;+\; \omega_\ell \frac{(1-n)^{1-\sigma_\ell}}{1-\sigma_\ell} \;+\; \aleph \;-\; \tau\bar{\tau}^{u} \;-\; \kappa_x H, \tag{14}$$

Four of these are the stylized model's and carry the same meaning here as in Section 3.2: γ the housing weight, θ the drug weight, so that $1-\gamma-\theta$ is the weight on consumption and the three sum to one; b the Stone-Geary endowment; and $\aleph$ the flow value of being alive. Equation (1) is the logarithmic case of this bundle with hours fixed.

The rest are new, and there are three of them. Leisure makes hours a margin, at weight ω_ℓ and curvature σ_ℓ over the $1-n$ hours not worked. SUD treatment costs utility, $\bar{\tau}^u$ in each period the household takes it up, $\tau = 1$. And aggregate homelessness, H, carries a disamenity, κ_x per unit, borne by every household, which is what makes homelessness an externality for everyone here rather than only a private misfortune; Section 5.1 takes it up. Outside the bundle, σ is the coefficient of relative risk aversion, and its inverse is the elasticity of intertemporal substitution.

4.3 Income, taxes and transfers

Gross income depends on age, employment, productivity and health,

$$y^{g}(j, y, e, d, m, n) = \begin{cases} \psi_R\, y & j \geq J_R \quad \text{(social security)} \\ n\, y\, \varepsilon(j)\, \rho_d^{\mathbf{1}\{d=1\}} \rho_m^{\mathbf{1}\{m=1\}} & j < J_R,\ e = 1 \quad \text{(employed)} \\ \psi_{ui}\, \min\{y\, \varepsilon(j),\ \bar{y}\} & j < J_R,\ e = 2 \quad \text{(first year of a spell)} \\ \zeta_{DI}\, y\, \varepsilon(j) & j < J_R,\ e = 3,\ m = 1 \quad \text{(disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{15}$$

taking the cases in the order they appear. Retirement income replaces a fixed share ψ_R of the productivity state with no SMI penalty, since social security is not means-tested on SMI. The base is y rather than $y\,\varepsilon(j)$,

since there is no work in retirement for the age profile to apply to. For the employed, $\varepsilon(j)$ is the deterministic age profile of efficiency and ρ_d and ρ_m[8] are productivity penalties applied to the SUD and to SMI respectively, *conditional on working*. Both are set to one in the benchmark, so the income equation is written in general and the two penalties are inactive in it; Table 5 says so and Section 5 says why.[9] The three employment states are work, $e = 1$, the first period of a spell out of work, $e = 2$, and non-employment thereafter, $e = 3$, so the last two cases are both unemployment and the disability case is the income of a household that is out of work and carries SMI. Unemployment insurance (UI) is paid in the first period of a spell only, at the replacement rate ψ_{ui} on an earnings base capped at average income $\bar{y}$, both as in equation (4). ζ_{DI} is the rate at which disability insurance (DI) replaces the income of an unemployed household with SMI, and that DI does not stop until the household recovers. It is set to zero, because that is what the vast majority of such households receive. Only about three in thirteen of them draw disability insurance at all; the other ten draw nothing. Some do receive it, so zero is an approximation, but it is the right one for a single number standing in for the whole group.

Labour income is taxed progressively with the Heathcote et al. (2017) function and capital income at a flat rate τ_k,

$$y^n = \tau_\ell \, (y^g)^{\xi} \; + \; (1 - \tau_k) \, r \, a, \tag{16}$$

with y^n denoting after-tax income, where ξ governs progressivity and τ_ℓ the level. Households can receive two transfers, both means-tested on gross income as the programmes themselves are. Food assistance pays the maximum allotment $\underline{c}\,\bar{y}$, with $\underline{c}$ expressed as a share of average income, less the fraction ϱ of means-tested income, which is the statutory rule and the same rule the stylized model uses, and the voucher covers rent above a tenant contribution of κ_r of gross earnings, the same κ_r as in Section 3.2.3,

$$F = \max\big\{0, \; \underline{c}\,\bar{y} - \varrho\,(y^g + ra)\big\}, \qquad Vch = \mathbf{1}\{\eta = 1\}\,\mathbf{1}\{h' \neq h_1\}\,\max\big\{0, \; p_r\,h' - \kappa_r\,y^g\big\}. \tag{17}$$

A voucher holder who becomes homeless does not lose the voucher: retention depends on income eligibility, not on housing, so the entitlement survives and simply pays nothing until the household rents again. Eligibility itself is set out in Section 4.9, equation (27), and the two policies' rules for withdrawing a voucher are set out with it.

4.4 The budget constraint

$$c + p_v v + a' \;=\; y^n + a \;-\; p_r\,h'\,\mathbf{1}\{h' \neq h_1\} \;+\; Vch + F, \qquad c > 0, \; v \geq 0, \; a' \geq 0. \tag{18}$$

The drug price is p_v and market rent is p_r, both as in Section 3.2.3. Note what is absent: treatment, which costs utility, and bequests from deceased households, which go to the government. How c, h, v and n are chosen out of it is with the rest of the optimisation, in Section 4.11. Rents are normalised to one and held fixed across every simulation.

4.5 Housing and the homelessness loop

Housing is the four point grid shown in (13). h_1 is literal homelessness, sleeping rough or in an emergency shelter. The three rental sizes are pinned at both ends and spaced by a curvature between them,

$$h_3 = h_2 + \Big(\tfrac{1}{2}\big(h_4 - h_2\big)^{1/\varsigma}\Big)^{\varsigma}, \tag{19}$$

so three parameters describe the grid: h_2 the smallest rental size, h_4 the largest, and ς the curvature that places the rung between them. At $\varsigma = 1$ the ladder is uniform and at $\varsigma > 1$ the middle rung sits closer to the bottom, which is what a housing size distribution with most households in the largest category requires.

Homelessness then enters three separate equations: survival (24), the employment transitions (26) and the SUD recovery hazard (21), through $\kappa_{\phi H}(j)$ and κ_{ueH} in Section 4.8 and μ_H in Section 4.6. It raises mortality, worsens both employment margins and lowers the chance of recovering from the SUD. SMI is the one thing it does not reach. The SMI chain runs on its own, with the reason given in Section 4.7.

[8] Flagged: ρ_d and ρ_m appear here. If either is discarded this equation changes.
[9] Flagged: keep the two penalties in the model at all, or take them out of the income equation?

4.6 Drug use, the SUD, recovery and treatment

A household either abstains or uses, with v chosen continuously. A clean household that uses acquires the SUD with the hazard of equation (2),

$$\Pr(d' = 1 \mid d = 0, v) = \pi_d(v) = 1 - e^{-\lambda_d v}, \tag{20}$$

In the baseline a clean household may initiate only up to an age J_E, the last age at which use can begin.[10] A household with the SUD recovers with probability

$$\pi_r(h', m, \tau) = \min\left\{1,\ \bar{\pi}_r\, \mu_H^{\mathbf{1}\{h'=h_1\}}\, \mu_m^{\mathbf{1}\{m=1\}}\, g^{\tau}\right\}, \tag{21}$$

with $\bar{\pi}_r$ the baseline annual chance of recovering while housed, clean of SMI and untreated, and three multipliers on it: $\mu_H < 1$, which means that recovery is harder in a shelter, $\mu_m < 1$, which implies that recovery is harder with SMI, and $g > 1$, which means that recovery is more likely in treatment.

μ_H is how housing acts on drug use here, and it is worth being exact about what that claim is and is not. Housing does not change what a household wants: flow utility is the same function of (c, h, v) housed or not. What housing changes is the chance of getting out of the disorder once it is carried, recovery in a shelter being μ_H times recovery while housed. Milby et al. (2005) randomised cocaine-dependent homeless adults into arms differing in housing and measured weeks of abstinence, so the ratio between arms identifies μ_H in our calibration.

Treatment τ is a *choice*, not a mechanical hazard, taken at the utility cost $\bar{\tau}^u$ in (14). It also costs the drug: a household in treatment must abstain in that period, $v = 0$ whenever $\tau = 1$, which is imposed as a constraint on (30) rather than priced. On the other hand, because it is not using, its mortality rate goes back to the baseline for that period.

g is the parameter through which its entire effect runs, so treatment has to be something a household decides to undertake and can decline.

4.7 Serious mental illness

m is an exogenous two state Markov chain representing SMI, whose prevalence $s_m(j)$ is a small share of adults, defined by substantial functional impairment, rather than any mental illness, which is roughly a quarter of adults and includes mild anxiety: a state that halves productivity cannot sensibly cover a quarter of the population.

The onset rate is a function of age. Writing $\varsigma_m(j)$ for its age profile, normalised to one at age 26, and ω_m for its level,

$$\text{onset}_m(j) = \omega_m\, \varsigma_m(j). \tag{22}$$

The profile is taken from the age profile of prevalence in NSDUH 2024; the level is calibrated, to the share of the population carrying SMI.

The transitions are then

$$\Pr(m' = 1 \mid m = 0, d) = \text{onset}_m(j) \cdot \iota^{\mathbf{1}\{d=1\}}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m, \tag{23}$$

where $\iota > 1$ is the multiplier on onset for a household carrying the SUD, which makes a SUD raise the hazard of SMI. That last channel runs in the direction the evidence supports: substance induced psychosis converts to schizophrenia or bipolar disorder at a pooled rate near a quarter (Murrie et al., 2020), and in the Danish registers any SUD predicts conversion to schizophrenia with a hazard ratio of 1.34, opioid use disorder 2.74 (Nielsen et al., 2017).

SMI then affects four objects: both employment margins through κ_{uem} in (26), mortality through $\kappa_{\phi m}$ in (24), the SUD recovery hazard through μ_m in (21), and disability income, for which eligibility is being out of work with SMI and the replacement rate is ζ_{DI}. It does not change the utility function.

[10] Without it there is little consequence at old ages to using: too little life remains for the disorder to damage, so the model produces too much of it.

The SMI chain does *not* depend on housing, and that is a finding in the data rather than an omission. In the At Home / Chez Soi randomised trial, 1,198 homeless adults with SMI across four cities, assignment to housing with intensive case management produced no detectable effect on symptom severity at any wave, while producing large effects on housing stability, its primary outcome. Making the channel housing dependent would mean adding a mechanism the best available trial looked for and did not find.

4.8 What the SUD, SMI, and homelessness cost

Death risk. Survival is

$$\phi(j, h', d, m, \tau) = \bar{\phi}(j)^{\,\kappa_{\phi H}(j)^{\mathbf{1}\{h'=h_1\}}\,\kappa_{\phi d}^{\mathbf{1}\{d=1\}\mathbf{1}\{\tau=0\}}\,\kappa_{\phi m}^{\mathbf{1}\{m=1\}}}, \tag{24}$$

with $\bar{\phi}(j)$ the baseline survival age profile and $\kappa_{\phi H}(j)$ an age-varying multiplier and $\kappa_{\phi d}$ and $\kappa_{\phi m}$ two that are not, raising the death hazard for homelessness, the SUD and SMI respectively. Each is a ratio of mortality *rates*, which is what the sources measure, so each multiplies the hazard rather than the death probability: survival is raised to their product. The two coincide while the baseline is small and part company where it is not, and raising survival to a power is bounded in $[0, 1]$ without a cap. The SUD multiplier is switched off in a period the household is in treatment, $\tau = 1$, which is why the survival function takes τ as an argument at all. The excess mortality of the disorder is overwhelmingly a mortality of use, poisoning above all, and treatment forbids use in the period it is taken, so a household that is not using does not carry it. I make this assumption because that is what the cause-specific evidence shows.[11] $\bar{\phi}(j)$ is the SSA period life table for ages 26 to 85, with the sexes combined by survivor weight, $q = [l^M q^M + l^F q^F]/[l^M + l^F]$.

The homelessness multiplier cannot be read off the data directly, and the reason is worth spelling out. Baggett et al. report a *standardised mortality ratio* (SMR) for homeless adults: deaths among the homeless at a given age divided by deaths in the general population at that age, which they publish by age band and which we write $RR(j)$. It cannot be used as $\kappa_{\phi H}(j)$ because the people it is measured on are not only homeless. A share $s = 0.217$ of them also carry a SUD (Gutwinski et al., 2021), which raises mortality by itself, so the published ratio is a mixture of two effects and attributing all of it to housing would overstate what housing does.

Two things are observed and one is solved for. Observed: $RR(j)$, Baggett's ratio, and s, the share of the homeless with a SUD. Solved for: $\kappa_{\phi H}(j)$. Both inputs are inputs to this step and not moments the model is asked to reproduce, so neither appears among the validation moments in Section 5.3.

$$\kappa_{\phi H}(j) = \max\left\{1,\ \frac{RR(j)}{(1-s) + s\,\kappa_{\phi d}}\right\}. \tag{25}$$

The denominator is the average mortality multiplier in Baggett's population: a share $1 - s$ of it carries homelessness alone and a share s carries the SUD as well. Dividing the observed ratio by it leaves the homelessness effect on its own.

$\kappa_{\phi d}$ is assigned from the literature on all-cause standardised mortality ratios for opioid use disorder, which report values between 6 and 10 (Degenhardt et al., 2011; Hser et al., 2017; Larney et al., 2020). Those are ratios of rates, which is why it enters as an exponent on survival rather than as a factor on the death probability.

$\kappa_{\phi m}$ carries no age argument because the data give it basically none. Olfson et al. (2015) give all-cause standardised mortality ratios for schizophrenia of 3.6, 3.4 and 4.0 across ages 20 to 34, 35 to 54 and 55 to 64. Basically flat.

Unemployment risk. Job finding and separation probabilities are

$$\pi^{ue} = \min\left\{1,\ \bar{\pi}^{ue}\,\kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{-\mathbf{1}\{d=1\}}\kappa_{uem}^{-\mathbf{1}\{m=1\}}\right\}, \qquad \pi^{eu} = \min\left\{1,\ \bar{\pi}^{eu}\,\kappa_{ueH}^{\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{\mathbf{1}\{d=1\}}\kappa_{uem}^{\mathbf{1}\{m=1\}}\right\}, \tag{26}$$

[11] In the data, the meta-analyses of mortality among people using extramedical opioids report overdose as the largest single contributor to the excess (Larney et al., 2020; Degenhardt et al., 2011), and in the Centers for Disease Control and Prevention (CDC) WONDER database drug poisoning alone accounts for about a fifth of all deaths at ages 25 to 44, which is the moment Section 5.3 reports the model against.

with $\bar{\pi}^{ue}$ and $\bar{\pi}^{eu}$ the baseline job finding and separation rates, respectively, and κ_{ueH}, κ_{ued} and κ_{uem} the multipliers for homelessness, the SUD and SMI, respectively, so each multiplier above one makes a job likelier to lose and regaining one less likely.

The child's productivity penalty. A parent with a SUD, or one who ends up in a shelter, lowers the productivity (y_{child}) its child is born with, through δ_d and δ_h respectively, as shown in (39) in Section 4.14: δ_d is the proportional cut to the child's productivity draw when the parent carries the SUD at the time of death, and δ_h the cut when the parent is homeless at the time of death. That is a risk to the household's own objective only to the extent that ω_c, the weight the parent places on its child's lifetime utility in (30), is positive. It is therefore the one risk here that carries an externality rather than a pure personal cost, which is what makes it a candidate for policy in a way that mortality and unemployment risk, both fully internalised, do not.

4.9 The voucher and the two policies

A household without a voucher draws one with probability π_V, the same lottery probability as in Section 3.2.3, if it is income eligible, and a household with one keeps it while eligible, so unlike Section 3.2.3 the voucher is persistent rather than redrawn each period. Eligibility is

$$y^g + ra \;<\; \kappa_V \, y^{\text{med}}. \tag{27}$$

with κ_V the eligibility cutoff expressed as a share of median gross income y^{med}. Under Housing First that is the whole rule. Under Treatment First the voucher is additionally lost with probability p_{det} by a household that uses drugs or has a SUD, and with probability p_{fp} by one who does not,

$$\Pr(\eta' = 1) = \Pr(\eta' = 1)^{\text{HF}} \times \begin{cases} 1 - p_{\text{fp}} & \text{if } (v = 0 \text{ and } d = 0) \text{ or } \tau = 1, \\ 1 - p_{\text{det}} & \text{otherwise.} \end{cases} \tag{28}$$

There are four cases and the rule covers all of them. A clean household that does not use meets the condition and faces only the false positive rate p_{fp}. A clean household that uses fails it and faces p_{det}. A household carrying the disorder that does not take treatment fails it and faces p_{det} as well. And a household carrying the disorder that does take treatment is abstaining, $v = 0$ by the restriction in Section 4.6, so it meets the condition and faces p_{fp}. There is no fifth case: treatment is available only when $d = 1$, and it forbids use, so no household is ever both in treatment and using.

This means that in contrast to Section 3.5, enforcement here is not perfect both ways. There, conditionality is perfectly enforced: a household that fails the condition always loses the voucher, and one that meets it never does. Treatment First deters everyone, so nobody is ever thrown off the rolls. Here detection is imperfect: some fail the condition even when they are clean (a false positive), and some pass even though they are using (a false negative). Neither rate binds in the benchmark, where no voucher is conditional, so both are set with the experiment in Section 7 rather than in the calibration.

4.10 The shocks, collected

Conditional on the state and the choices, next period's state is drawn from five independent transitions. Collecting them in one place:

Productivity y. Fixed over the life cycle, so $y' = y$. I do this for two reasons. First, it is what keeps the child's penalty (39) a shift along a single scalar rather than a shift across the nodes of an age-varying discretisation, which significantly reduces the computation time to manageable levels. Second, it still captures the key mechanisms, because unemployment is the income risk that links income to the SUD and the SUD to income, and it is retained in full.

Employment e. A household that loses their job receives insurance with certainty for one year. With π^{ue} and π^{eu} from (26), it gives the following transitions:

$$e=1 \;\longrightarrow\; \begin{cases} e'=1 & \text{w.p. } 1-\pi^{eu} \\ e'=2 & \text{w.p. } \pi^{eu} \end{cases} \qquad e\in\{2,3\} \;\longrightarrow\; \begin{cases} e'=1 & \text{w.p. } \pi^{ue} \\ e'=3 & \text{w.p. } 1-\pi^{ue}, \end{cases} \tag{29}$$

State 2 is reachable only from employment and never from itself, so a household is in it for the first year out of work and then moves to state 3, which pays nothing, from the second year on. One year is if anything generous: most states pay for twenty six weeks only.

The SUD d. By (20) if $d=0$ and (21) if $d=1$, conditioning on the chosen v, h' and τ.

Mental illness m. By (23), conditioning on d.

The voucher η. By (27) and (28), conditioning on next period's eligibility and, under Treatment First, on the chosen v, the status d and the treatment choice τ.

Write $\Gamma\big(s' \mid s,\, n, h', v, \tau\big)$ for the product of these five, which is the object the expectation in (30) is taken against.

4.11 The household's dynamic problem

Putting it together, for $j=1,\dots,J$ and with $V(J{+}1,\cdot)$ given below,

$$\begin{aligned} V(j,a,y,e,d,m,\eta) = \max_{\{n,\,h',\,v,\,\tau,\,a'\}} \Bigg\{ & \frac{\big[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\big]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} - \kappa_x H \\ & + \;\beta\,\phi(j,h',d,m,\tau)\int_{\mathcal{S}} V(j{+}1)(s')\,\Gamma\big(\mathrm{d}s' \mid s,n,h',v,\tau\big) \\ & + \;\beta\,\big[1-\phi(j,h',d,m,\tau)\big]\,\omega_c\Big[\mathbb{E}\,V\big(1,0,y_{\text{child}},e',0,0,0\big) \\ & \qquad - \;\mathbb{E}\,V\big(1,0,y_{\text{base}},e',0,0,0\big)\Big] \Bigg\} \end{aligned} \tag{30}$$

subject to the budget (18), the income definitions (15) to (17), and the choice set of dwellings (13).

Four restrictions apply to the choices. Treatment is available only to a household that carries the disorder, $\tau=0$ whenever $d=0$, and it requires abstention, $v=0$ whenever $\tau=1$. Hours are zero for anyone not working, $n=0$ whenever $j\geq J_R$ or $e\neq 1$. Borrowing is not allowed, $a'\geq 0$, and $c\geq F$, so that the food stamp payment cannot be saved, since it is paid in kind.

The terminal condition is $V(J{+}1,\cdot)=0$ with $\phi(J)=0$, so at the last age the child is the whole continuation. The integral runs over next period's state $s'=(a',y,e',d',m',\eta')$; the second line of (30) is the household's own continuation, and the third is the child's expected lifetime utility, entered at age one with zero assets and with its own employment status e' drawn as any newborn's is.

A household that does not survive from j to $j+1$ leaves a child. The continuation is therefore two terms, each discounted: the survival-weighted value of the household's own next period, and the death-weighted value of what its own state costs that child, with y_{child} as defined in (39). What that buys is a second route by which the household's own decisions, to use and to go unhoused, reach its child. $\phi(j,\cdot)$ itself depends on the dwelling chosen, so the housing decision moves the weight on the two terms directly. The drug does not move it within the period, because ϕ takes the disorder as a state and v only raises the chance of carrying it *next* period. The drug reaches the child through the continuation instead, which is where the first-order condition (34) picks it up.

The third line is a *difference* and not a level, and the distinction is not cosmetic. Written as a level, ω_c times the child's utility, the term is worth more to a household the likelier it is to die, because it is the death

branch that carries it. A SUD raises the death hazard several fold, so a level would pay most to exactly the households the paper is about, and it would do so for reasons that have nothing to do with the child.

Subtracting the value of a child born to a parent who was clean and housed when they died removes that. The term is zero for a household in the reference state and negative otherwise, so what enters the decision is only the loss the parent's own state imposes on its child, weighted by the chance that child is born at all.

At $\omega_c = 0$ the damage a household does to its child is entirely external; at $\omega_c > 0$ the household internalises a share ω_c of it, and only $1 - \omega_c$ remains external. Because the term is a difference, ω_c is the weight on that damage and nothing else. Since the child's productivity depends on whether the parent carried a SUD or was homeless, and both depend on v and h', the child term in (30) is what puts the child inside the parent's drug and housing decision.

The chain down the lineage is priced geometrically: the child's value $V(1, \cdot)$ on the third line contains its own ω_c-weighted child term, so generation k enters the parent's problem at ω_c^k and $1-\omega_c^k$ of it stays outside. What is meant throughout is the lineage.

How the choices separate, and where they do not. The problem looks like five choices at once, but some of it separates. Hours, the dwelling and saving can be taken as given while the drug and consumption are split out of what remains, and that split is where the whole of the model's substance sits: it is the one margin on which the household weighs a flow pleasure against a change in its own future state.

Taking hours, the dwelling and next period assets as given, what is left to divide between consumption and the drug is

$$x \;=\; y^n + a \;-\; p_r\, h'\, \mathbf{1}\{h' \neq h_1\} \;+\; Vch + F \;-\; a', \tag{31}$$

so the budget (18) reads $c + p_v v = x$ and the whole of the remaining problem is one number: how to split x between c and v, since either determines the other.

Were the drug an ordinary good, one whose only consequence is the flow utility it delivers this period, that split would follow from flow utility alone. It does not, because v enters the transition as well as the utility: by (20) another unit of the drug also buys a higher chance of waking up next period with the disorder. So there are three terms in the first-order condition for v, not two,

$$\underbrace{\frac{\theta}{b+v}\,\Lambda}_{\text{value of another unit}} \;=\; \underbrace{\frac{p_v\,(1-\gamma-\theta)}{c}\,\Lambda}_{\text{the consumption it displaces}} \;+\; \underbrace{\lambda_d e^{-\lambda_d v}\Big[\,\mathbb{E}V(\cdot \mid d'=0) - \mathbb{E}V(\cdot \mid d'=1)\,\Big]}_{\text{the risk it adds}}, \tag{32}$$

where $\Lambda = \big[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\big]^{1-\sigma}$ is the Cobb-Douglas bundle of (14). It multiplies the first two terms and not the third, because the third is a difference of continuation values and is therefore already in utils. Were it absent, Λ would divide out of the condition entirely and the split would reduce to the usual Cobb-Douglas one; it is the third term that stops it cancelling, and that is why the marginal utility level matters here and not in an ordinary two good problem. That third term is the marginal hazard $\pi_d'(v) = \lambda_d e^{-\lambda_d v}$ times what the disorder costs, the gap between the two continuation values. That gap is where the externality on the child enters the drug decision, and it is why there is no fourth term.[12]

There are two cases in which the third term is zero, and in both the household faces no such trade-off: one that abstains, where $v = 0$, and one that already carries the disorder, whose recovery hazard (21) does not depend on how much it uses. For both the condition is the Cobb-Douglas split in closed form,

$$c = \max\Big\{F,\; \frac{1-\gamma-\theta}{1-\gamma}\big(x + p_v b\big)\Big\}, \qquad v = \max\Big\{0,\; \frac{x-c}{p_v}\Big\}, \tag{33}$$

whose corner is the abstention decision: a household buys nothing when the Stone-Geary endowment already exceeds what it wants, which is how $b > 0$ deters use and is the threshold of Section 3.2.4. The outer maximum on c is the food stamp payment F. Where it binds the Cobb-Douglas split no longer describes the allocation:

[12] Each continuation value carries its own child term, at $d' = 0$ and at $d' = 1$, so the difference between them already contains the ω_c share of what the disorder costs the child alongside what it costs the household itself. A fourth term would be needed only if using raised the chance the child's term is realised within the period, and it does not: ϕ depends on the disorder as a state, not on how much is bought today.

consumption is held at F and the drug takes what the budget leaves, so the floor changes behaviour and not only welfare. Where the third term is present the answer lies between (33) and abstention, since risk can only push towards less, so it is found by a one dimensional search on that interval.

Hours then solve

$$u_c(c)\,\frac{\partial x}{\partial n} \;=\; \omega_\ell (1-n)^{-\sigma_\ell}, \tag{34}$$

and the clawbacks in (17), at the statutory rate ϱ for food stamps and κ_r for the voucher, make $\partial x/\partial n$ jump *up* at the hours where each transfer runs out, because the clawback stops: an extra hour is worth more once the transfer is gone, so the household works more and takes less leisure. That is the welfare trap. It is the transfers, and the programme eligibility cutoffs that create it, that produce this, and both policies have them.

4.12 The government

The government pays social security, unemployment insurance, disability insurance, food stamps and vouchers, and supplies a public good $\bar{G}$ that does not enter private consumption and is held fixed across simulations, so that policies are compared at a constant level of it. It finances this with the two taxes in (16) and the estates of the deceased, with the level τ_ℓ clearing the budget in every simulation,

$$\underbrace{\textstyle\sum y^g - \tau_\ell \sum (y^g)^{\xi} + \tau_k \sum ra + \text{estates}}_{\text{revenue}} \;=\; \underbrace{SS + UI + DI + F + Vch + \bar{G}}_{\text{outlay}}. \tag{35}$$

Here a policy that houses more households spends more, τ_ℓ rises to pay for it, and the higher tax lowers the after-tax return to an extra hour: it enters (34) through $\partial x/\partial n$, which is net-of-tax as in (16), so hours fall and saving with them, for everyone, including the households the programme never reaches, and it runs against Housing First.

4.13 The drug market

Drug production has convex cost $C(S_v) = \frac{\nu_1}{\nu_2} S_v^{\nu_2}$, where S_v is the aggregate quantity. Supply satisfies

$$p_v = \nu_1 S_v^{\nu_2 - 1}, \qquad S_v = \int v(s)\,\mathrm{d}\Phi(s), \tag{36}$$

so that $1/(\nu_2 - 1)$ is the price elasticity of supply. ν_1 is normalised by setting its value such that the quantity supplied equals the quantity demanded at the benchmark price of one, and then held fixed across experiments. The price itself is not fixed: it must adjust across simulations to clear the drug market. The supply elasticity, its range and the evidence behind them are set in Section 5.

4.14 The law of motion and newborns

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi_{j+1}(\mathcal{S}') \;=\; \phi(j,\cdot) \int \Gamma\big(\mathcal{S}' \mid s,\, n_j(s), h'_j(s), v_j(s), \tau_j(s)\big)\,\mathrm{d}\Phi_j(s) \qquad \text{for every measurable } \mathcal{S}', \tag{37}$$

together with the newborn inflow

$$\Phi_1(\mathcal{S}') \;=\; \int \big[1-\phi(j,\cdot)\big]\,\mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_j(s) \;+\; \int \mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_J(s), \tag{38}$$

where $\mathcal{N}$ places a newborn according to (39), so that the mass of newborns equals the mass of the dead exactly and the two equations together conserve the population.

Here $\phi(j,\cdot)$ is evaluated at the household's own $(h'(j,s), d, m)$. The mass that does not survive, plus the mass at $j = J$, is replaced by newborns at $j = 1$ with zero assets, clean and without a voucher, carrying

SMI with the prevalence observed at age one rather than none of it, and subject to the same unemployment shocks as anyone else, whose productivity is drawn from the unconditional distribution Ω_y of permanent productivity across households scaled by the transmission penalties δ_d for a child whose parent carried the SUD when it died and δ_h for one whose parent was homeless then,

$$y_{\text{child}} \;\sim\; \Omega_y \;\text{ scaled by }\; (1-\delta_d)^{\mathbf{1}\{\text{parent had a SUD when it died}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless when it died}\}}. \quad (39)$$

The evidence behind both penalties is set out in Section 1 and placed against the literature in Section 2, so I focus here only on what functional form it takes: a parent's disorder or homelessness at death scales the child's productivity draw, by δ_d and δ_h respectively.

4.15 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\left\{\{V_t(j,\cdot)\}_{j=1}^{J}\right\}_{t=0}^{\infty}$;
- policy functions $\left\{\{n_t(j,\cdot),\, h'_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\right\}_{t=0}^{\infty}$;
- distributions $\left\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\right\}_{t=0}^{\infty}$; and
- prices, policies and aggregates $\left\{\tau_{\ell t},\, p_{vt},\, \bar{y}_t,\, y_t^{\text{med}},\, H_t\right\}_{t=0}^{\infty}$,

such that, at every date t,

1. the policy functions solve (30) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;
2. $\Phi_{t+1}(j{+}1,\cdot)$ follows from $\Phi_t(j,\cdot)$ by the law of motion (37) together with the newborn rule (39) and (38);
3. the government budget (35) holds at $\tau_{\ell t}$;
4. the drug market clears at p_{vt}, so (36) holds with aggregate drug demand $S_{vt} = \int v_t(s)\,\mathrm{d}\Phi_t$;
5. the public good is supplied at its assigned level, $G_t = \bar{G}$;
6. $\bar{y}_t$, median gross income y_t^{med} and the homelessness rate H_t are the aggregates implied by Φ_t.

A stationary equilibrium is the special case in which all of these are constant and $\Phi_{t+1} = \Phi_t$, which is what the calibration and the long-run policy comparisons report. The sequence version is stated for completeness, and because the transition between the benchmark and some of the long-run policy options is considered as well, in Section 3.7.

5 Calibrating the quantitative model

The model has fifty six parameters, some of which vary in value by age. Thirty five are calibrated externally, from data or from the literature, and twenty one are calibrated internally. Four further parameters govern drug testing and detection; they do not play a role in the benchmark, where nothing is conditional, but they do under Treatment First, which is where they are set. The calibrated parameters are split across the two tables of this section: thirty are the main story of this paper and its policy analysis (Table 5, thirteen internally and seventeen externally calibrated), and twenty six are standard (Table 6, eight internally and eighteen externally).

The tables carry the parameter, its description, the moment and source used to fix its value, and the value itself. The text alongside each table adds what a table has no room for: the intuition for what each parameter does, and how its moment identifies it. `data/DATA_APPENDIX.md` in the replication package documents every estimate in full, naming the dataset behind each target, its year and universe, and the computation that produces it. What follows gives the moment, its source and its value, and nothing else. The package carries the extracts themselves alongside the code that reads them, so nothing has to be downloaded.

5.1 The thirty parameters that carry the story of this paper

Twenty six of the fifty six are common to most life-cycle models, whereas the rest are specific to this paper, and all thirty of those carry its main story and policy analysis. They are set out in Table 5, with the

parameter, its description, and the moment and source used to fix its value, which is given in the last column. They are discussed one at a time below because they are the parameters a reader may want to argue over, and each is taken in the order of the table.

The comparison between Housing First and Treatment First has a shape, and every entry below is read against it, so it is worth setting out this comparison once and in full before the entries begin.

Three things decide the ranking between Housing First and Treatment First. The **private value** is what a household itself gains from being housed and from not carrying a disorder: consumption, a dwelling rather than a shelter, and above all the years of life that homelessness and the disorder take away.

The **externality** is what lands on somebody else: on the child, who inherits not the disorder or homelessness but a worse productivity draw associated with the parent's SUD and with the parent's homelessness, which δ_d and δ_h price separately. This means that the child is likelier in turn to be poor, to use, to acquire a disorder and mental illness as a consequence, to be unhoused and to die earlier. The parent internalises only ω_c of the part that falls on its child, and ω_c is small, which is what leaves something for a policy to correct. Another externality lands on everybody, through the disamenity of aggregate homelessness.

Lastly, the **distortion** is what the programmes cost in behaviour, and it comes from two places. The programmes create distortions: assistance is means-tested, eligibility depends on income and assistance is clawed back at a rate of κ_r, so a recipient who works more loses eligibility or part of the transfer, which requires even higher taxes on everyone else to fund even larger programmes. On top of that, assistance is paid for by distortionary taxes, which lower the after-tax return to an hour for households neither policy ever reaches.

Housing First is redistributionary, reduces two externalities, but amplifies a different externality. A child of a housed parent does not carry the penalty δ_h prices, and aggregate homelessness falls, which every household values through κ_x. However, it magnifies one at the same time, because assistance that arrives whatever a household does insures it against the consequences of using, this creates moral hazard, and so the SUD rate rises and reaches the child through δ_d.

Treatment First does not redistribute, mitigates an externality, but strengthens two others. Curing the parent is how the externality on the child's worse draw, δ_d, is avoided. But conditioning assistance means some households do not receive it and end up homeless, especially upon implementation, and over a transition to a new long-run equilibrium it amplifies the two externalities Housing First reduces. The transition matters on its own account: a household that wants to stop cannot do it immediately, and the transition is costly while it lasts, which is why the policy analysis reports the transition and not only the long-run steady states.

And general equilibrium offsets some of the effects. First, the drug price is endogenous, so curing disorders lowers drug demand, which lowers the equilibrium price, and makes using cheaper for those still deciding. Second, when the government spends more on either policy and the tax rises, hours fall, the tax base shrinks and the tax has to rise again. Third, Treatment First has an offset of its own in the other direction: conditionality induces more households to seek treatment to help them recover more quickly, and this helps reduce the externality on newborns during the transition. And because κ_x is levied on the aggregate headcount, any movement in homelessness feeds back into everyone's utility.

For every parameter the things worth knowing are what happens to the private value, to the externality and, where it applies, to the distortion, *as its value rises or falls*, and each entry says so in a line before giving the moment. The middle blocks are where the two move together: because the child inherits a higher chance of landing in exactly the states those parameters price, anything that raises the private cost of using or of being unhoused raises the externality as well.[13] Both policies work on those blocks, from opposite ends.

Who uses drugs. These four set participation and, more importantly, *who* participates. Neither policy acts on them directly; they decide how much there is for a policy to act on.

θ, the weight on drugs in utility. How much a household wants the drug at all, against consumption and housing. It is the parameter that determines the amount spent on drugs when the household consumes a strictly positive amount of it. As it rises use rises on both the extensive and intensive

[13] Flagged: Table 5 below carries a row for ρ_d and a row for ρ_m, both reading "Set to one; the extensive margin carries it". Both rows go if the penalties are discarded, and the parameter counts in this section's opening move with them.

Table 5: The thirty parameters that carry the story of this paper

Parameter	Description	Set	Moment and source	Data	Model	Value
Who uses drugs						
θ	Weight on drugs in utility	Int.	Drug spending over income, Midgette et al. (2019)	16.2%	—	—
b	Whether a household uses	Int.	Drug participation, NSDUH 2024	8.90%	—	—
λ_d	SUD hazard per unit used	Int.	Adult SUD rate, NSDUH 2024	3.68%	—	—
$\kappa_{\phi d}$	Increased mortality with a SUD	Ext.	Mortality ratios, Hjorthøj et al. (2023)	6 to 10	n.a.	8.00
$\aleph$	Flow value of being alive	Int.	Value of a statistical life, Viscusi and Aldy (2003)	\$9m	—	—
What using and being homeless cost, and the voucher						
ρ_d	Earnings lost to a SUD	Ext.	Set to one; the extensive margin carries it	0.50	n.a.	0.50
κ_{ued}	Employment multiplier, SUD	Int.	Unemployment rate given a SUD, NSDUH 2024	16.25%	—	—
h_1	Service flow of homelessness	Int.	Homelessness rate, HUD 2023	0.399%	—	—
$\kappa_{\phi H}(j)$	Increased mortality when homeless	Ext.	Mortality ratios by age, Baggett et al. (2013)	9.1 to 1.1	n.a.	4.41 to 1.09
κ_{ueH}	Employment multiplier, homeless	Ext.	Desmond and Gershenson (2016)	0.80	n.a.	0.80
$\underline{c}$	Food stamp maximum allotment	Int.	Food stamps over labour income, USDA	0.42%	—	—
$\bar{\pi}_V$	Chance of drawing a voucher	Int.	Recipients over eligible, HUD 2023	26.6%	—	—
Mental illness chain						
ι	SUD multiplier on SMI onset	Ext.	Danish registers, Nielsen et al. (2017)	2.00	n.a.	2.00
λ_m	Chance of recovering from SMI	Ext.	Ten year recovery rate, Åsbø et al. (2022)	50%	n.a.	0.067
ω_m	Level of the SMI onset profile	Int.	SMI share of the population, NSDUH 2024	4.82%	—	—
ζ_{DI}	Disability insurance, unemployed	Int.	SMI share of homeless, Gutwinski et al. (2021)	25%	—	—
ρ_m	Earnings lost to SMI	Ext.	Set to one; the extensive margin carries it	1.00	n.a.	1.00
κ_{uem}	Employment multiplier, SMI	Int.	Employment given SMI, Mechanic et al. (2002)	30%	—	—
μ_m	SUD recovery multiplier, SMI	Ext.	Dual diagnosis relapse, Bradizza et al. (2006)	0.82	n.a.	0.82
$\kappa_{\phi m}$	Mortality multiplier, SMI	Ext.	Olfson et al. (2015)	2.50	n.a.	2.50
Recovery and treatment						
$\bar{\pi}_r$	Baseline SUD recovery rate	Ext.	Ten year remission rate, Vaillant (1995)	0.025	n.a.	0.025
μ_H	Recovery multiplier when homeless	Int.	Weeks abstinent, Milby et al. (2005)	1.473	—	—
g	Treatment effect on recovery	Ext.	Odds ratio, Milby et al. (2005)	1.35	n.a.	1.35
$\bar{\tau}^u$	Utility cost of treatment	Int.	Treatment take up, NSDUH 2024	24.5%	—	—
Externalities and distortions						
ω_c	Weight on the child	Int.	Use rise with altruism off, NSDUH 2024	18.3%	—	—
δ_d	Child's earnings, parent SUD	Ext.	Earnings lost in utero, Nilsson (2017)	0.24	n.a.	0.24
δ_h	Child's earnings, homeless	Ext.	Earnings lost to poverty, Chetty et al. (2016)	0.31	n.a.	0.31
κ_x	Disamenity of homelessness	Ext.	Willingness to pay, Loubière et al. (2020)	0.20%	n.a.	0.002
σ_ℓ	Curvature on leisure	Ext.	Frisch elasticity, Chetty (2012)	0.50	n.a.	3.00
Drug market						
ν_2	Drug supply curvature	Ext.	Supply elasticity, Mace et al. (2020)	1.46	n.a.	1.68

Note: n.a. in the Model column marks a parameter that is set from outside rather than calibrated inside the model. For those the model produces no internal target to compare with the data, so the cell reads n.a. rather than a dash, which keeps them distinct from the internally calibrated parameters. For two parameters the source measures an elasticity while the model is written in a curvature, so the Data column gives the elasticity and the Value column gives the curvature that delivers it: σ_ℓ, where a Frisch elasticity of 0.5 needs a curvature of 3.00, and ν_2, where a supply elasticity of 1.46 needs a curvature of 1.68 through $1/(\nu_2 - 1)$.

margins, so the private cost and the externality both rise and there is more for either policy to act on. It is calibrated internally to drug spending over income across users, 16.2 percent, from retail drug spending in Midgette et al. (2019) and the incomes of adults aged 26 and over who use in NSDUH 2024. Section 1 of the data appendix builds it from those two.

b, whether a household uses at all. The marginal value of the drug goes as $\theta/(b+v)$, so a larger b makes the first units worth less and a household with little to spend buys nothing. As b rises use falls, so the private cost and the externality both fall, and housing assistance is less likely to be translated into more spending on drugs: b affects the degree of moral hazard. Its value is chosen so the model generates a drug participation rate of 8.90 percent, the NSDUH 2024 figure for adults aged 26 and over.

λ_d, the hazard per unit used. This is what makes using *riskier*, and it is the step that turns drug consumption into a disorder. Raise λ_d and both the private cost and the externality rise, and the moral hazard costs more, since it is λ_d that converts an extra voucher under Housing First into a higher SUD rate that Treatment First then attempts to cure. Its value is chosen so the model generates the share of adults carrying a SUD, 3.68 percent among those aged 26 and over in NSDUH 2024.

$\kappa_{\phi d}$, increased mortality with a SUD. This is the parameter the whole paper turns on. Mortality truncates the value of life for a user, which hits households with the highest life values hardest and thus reverses the income gradient, making the poor the users rather than the rich. When it rises, the private

cost rises, and the externality rises with it, since a child born to a parent who died with a SUD has a lower value of life. Both policies reduce these deaths, Treatment First by curing and Housing First too, because recovery is less likely when homeless. It is set externally, at the midpoint of the standardised mortality ratios for SUD in Hjorthøj et al. (2023), which run six to ten.

ℵ, the flow value of being alive. ℵ is the other half of the mortality mechanism: $\kappa_{\phi d}$ sets how much a SUD raises the chance of dying and ℵ sets what the life being lost is worth. Increase it and the private value of every life either policy saves rises, and use falls, so the externality falls. Its value is chosen so the model generates a value of a statistical life of $9m, from Viscusi and Aldy (2003).[14]

What using and being homeless cost, and the voucher. Both policies work on this block, from opposite ends: each parameter says how bad a state is, so each sets what a voucher is worth to a household in that state and what being cured of that state is worth. The voucher itself closes the block. This is because it is the instrument that acts on all of them. Because the child inherits a higher chance of landing in these states, everything here moves the private value and the externality together.

ρ_d, earnings lost to a SUD while still working. [15] It is set to one, so the extensive margin carries the earnings loss. It reduces the value of life when a household acquires a SUD, since the earnings it costs would have been enjoyed for the rest of that life. What lets an *employed* household be poor enough to be priced out of housing is h_2, the price of the cheapest dwelling. As ρ_d rises the penalty shrinks, so the private cost falls and, because the impact on the child is less harmful, the externality declines. These earnings are lost whether the household is housed or homeless, so treatment affects them and housing does not.[16]

ρ_d is what makes the disorder costly at every income rather than only through the job, so it is what decides how much of the homeless population is working: κ_{ued} carries the disorder into unemployment, and this carries it into poverty even when the household keeps its job. Its value is chosen so the model generates the 50 percent of homeless adults who are employed, from HUD's 2023 PiT count and the Treatment Episode Data Set, Admissions (TEDS-A) 2022. Raise it and the penalty shrinks, so a working household with the disorder can afford housing again and that share falls.

κ_{ued}, the employment multiplier for a SUD. This parameter decides how much less likely regaining employment is, and how much more likely losing it is, with a disorder. Raise it and the private cost rises, the externality rises with the child's draw, the tax base narrows since the household does not work, and the income tax rises because the household receives unemployment insurance. Its value is chosen so the model generates the unemployment rate given a SUD, 16.25 percent in NSDUH 2024.

h_1, the service flow of homelessness. It sets how bad the outside option is, and the value of being alive is measured against that outside option: a household with little to lose from dying is one for which the mortality cost of using is small. When h_1 rises, a shelter is less bad, so the private cost of homelessness falls, a voucher is worth less, and the externality on the child likelier to be unhoused falls with it. Its value is chosen so the model generates the homelessness rate, 0.399 percent of households in HUD's 2023 PiT count.

$\kappa_{\phi H}(j)$, mortality when homeless. Increase it and the private cost of being unhoused rises and thus so does the private value of a voucher. Baggett et al. (2013) give race adjusted mortality ratios for homeless adults of about 9.1 at ages 25 to 44, falling to 1.1 at 65 to 84. Those are totals for a population most of which also carries a SUD, so it is set externally age by age net of that composition, which is why it is derived jointly with $\kappa_{\phi d}$ rather than read off directly, as set out in Section 4.8 at (25).

κ_{ueH}, the employment multiplier for homelessness. This parameter decides how much more likely a homeless household is to lose its job and how much less likely it is to regain it. Housing First mechanically recovers earnings while Treatment First does not. As this parameter's value rises the private cost and the externality rise together, which makes it the clearest case in the table of a parameter loading

[14] No household has a negative value of life in equilibrium, so death is never an optimal outcome.

[15] Flagged: keep ρ_d or take it out?

[16] One subtlety is worth stating, because it may seem like a contradiction with Section 3.1. There the proportional income penalty was shown to matter little for *who* uses. That is a statement about the calibrated model, not the parameter: raising ρ_d with b held fixed does change who uses, but b is recalibrated to the participation rate each time the penalty changes, and once it is, the composition of users is back where it started.[17]

Housing First's side. Moreover, it reduces the tax burden since these households are more likely to be employed and off the unemployment insurance roll. It is set externally at 0.80, from Desmond and Gershenson (2016).

$\underline{c}$**, the food stamp maximum allotment.** This parameter determines how low consumption can go, which equals the transfer food assistance pays at zero income. When it rises, the private cost of every bad state falls, so a voucher is worth less at the margin and the externality on a child likelier to reach those states falls too, while the distortionary taxes that fund it rise. It is calibrated internally to food stamp spending over total labour income, 0.42 percent, from the United States Department of Agriculture (USDA).

π_V**, the chance of drawing a voucher.** This is the policy dial itself. Increase it and the private gain rises and two externalities fall mechanically: the child's penalty for growing up unhoused and the aggregate homelessness disamenity, while the third externality rises through the SUD rate and the distortion rises with the higher bill to finance it: the whole trade in one parameter. Its value is chosen so that only 26.6 percent of the income-eligible population receives a voucher, as in HUD's Picture of Subsidized Households, 2023.

Mental illness chain. This block determines how much further Treatment First reaches over and above the disorder it treats: a SUD raises the chance of mental illness, mental illness lowers employment and raises death, and the child is susceptible to all of this too. Everything here therefore moves the private cost and the externality in the same direction, and the first three decide whether the chain exists at all.

ι**, the SUD multiplier on SMI onset.** The hinge of the whole block. Without it mental illness is exogenous to everything a policy does and treating SUD cannot touch it. As it rises a SUD causes more mental illness, so the private cost of using rises, and the externality rises further, because the child of a user now inherits a higher chance of illness that is a consequence of the productivity penalty. Treatment First's return rises as ι rises, because curing the disorder now prevents some cases of SMI too. It is set externally at two, from the Danish registers in Nielsen et al. (2017).

λ_m**, the annual chance of recovering from SMI.** This parameter governs how long an episode of mental illness lasts: it is the hazard of leaving the SMI state each year, so it sets the persistence of the illness rather than how many households ever acquire it. Raise it and mental illness is less persistent, so the private cost falls, because fewer years are spent with SMI, and the externality falls with it, since a child who is likelier to acquire a SUD and through it SMI has a higher chance of recovery and a higher value of life. It is set externally at 0.067, the annual hazard that gives a 50 per cent chance of recovery within ten years, the ten-year clinical recovery rate reported by Åsbø et al. (2022).

ω_m**, the level of the SMI onset profile.** This parameter governs how many households ever acquire mental illness: it scales the age profile of onset up or down without changing its shape, so it fixes the size of the SMI population rather than who reaches it or when. Raise it and both the private cost and the externality rise, since a child is likelier to reach a state that costs earnings, employment and years of life. It is calibrated internally, to the share of the population carrying SMI in NSDUH 2024, which is 4.82 percent.

ζ_{DI}**, disability insurance for the unemployed.** This parameter sets the floor under one of the worst states in the model, being ill and out of work, so it decides how far such a household can fall. Raise it and the private cost of that state falls, and with it the externality on a child likelier to reach it, at the price of a higher tax on everybody else; lower it and more households with SMI fall to the same zero income as everyone else out of work, so more of them lose their housing. It is set externally to zero, because the vast majority of unemployed households with serious mental illness receive no disability insurance: roughly three in thirteen draw it and the other ten draw nothing. A positive value would give all thirteen a floor that ten of them do not have. Some households do receive DI, so zero is an approximation rather than a description, but it is the closer of the two available to a group that mostly receives nothing. The SMI share of the homeless is then reported with the untargeted moments rather than matched.

ρ_m**, earnings lost to SMI.** [18] The counterpart of ρ_d for mental illness, and it reaches the comparison

[18] Flagged: keep ρ_m or take it out?

through ι: a household carrying the disorder faces onset at ι times the baseline rate, so treatment reduces mental illness in a roundabout way and recovers these earnings for households that never fall ill. When it rises, the penalty shrinks, so the private cost of illness falls and, because the child's draw improves with it, the externality falls too. It is set to one in the benchmark, for the same reason as ρ_d and for one of its own. Working with SMI would otherwise pay less than not working with it, because an unemployed household with the illness draws disability insurance at ζ_{DI}, so the penalty inverted the return to work at the values that fitted. The income of the homeless relative to the average is reported with the untargeted moments instead.

κ_{uem}**, the employment multiplier for SMI.** This parameter is the same as κ_{ued} but acts on SMI: it determines how much more likely a household with SMI is to lose its job and how much less likely it is to regain it. Increase it and the private cost rises, because being ill costs the job, and the externality rises with the child's draw. Raising it narrows the tax base, because the household does not work, and raises spending, because it receives disability insurance, both of which increase the distortion by raising the income tax rate on everybody else. A child of a SUD-treated parent is less likely to end up with SMI, which raises its earnings and lowers its chance of being unhoused, so part of Treatment First's return appears in the health of the children of those treated for a SUD rather than in the treated household's own drug use. Its value is chosen so the model generates the employment rate given SMI, about 30 percent, the midpoint of the range for high-disability diagnoses in Mechanic et al. (2002).

μ_m**, the SUD recovery multiplier for SMI.** This parameter is the second direction of the chain: not only does a SUD increase the chance of SMI, but SMI also means that the SUD is less likely to be cured. As it falls the disorder is harder to recover from, so the private cost of acquiring one rises and the externality rises with it, and Treatment First has more to overcome in exactly the households it most wants to reach. It is set externally at 0.82 from the dual diagnosis relapse rates in Bradizza et al. (2006), whose review covers people carrying both a SUD and a mood, anxiety, schizophrenia-spectrum or personality disorder. Section 8 of the data appendix takes their relapse rates to the multiplier.

$\kappa_{\phi m}$**, the mortality multiplier for SMI.** The chain's last link, and the channel that makes it matter for welfare, not through income. As it rises deaths among the mentally ill rise, so the private cost rises, and the externality rises with it, because a child born to a parent with a SUD is more likely to acquire a SUD and through it SMI, and therefore more likely to die younger. This is how a SUD affects mortality indirectly, through SMI, rather than only by the direct route. It is set externally at 2.5, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

Recovery and treatment. Both policies work through parameters in this block, since recovery is what Treatment First buys directly and what Housing First buys by housing people. The first two set how much recovery there is to buy and the last two how much of it a policy can reach.

$\bar{\pi}_r$**, the baseline SUD recovery rate.** This parameter is the hazard of getting out of the disorder for a housed household, which is the level everything else in this block scales. As it rises disorders end sooner on their own, so the private cost of acquiring one falls and the externality falls with it, and there is less for either policy to buy. It is set externally from the ten-year remission rate in Vaillant (1995), which gives an annual hazard of 0.025.

μ_H**, SUD recovery when homeless.** The value of this parameter determines how much less likely recovery from the disorder is for an unhoused household than a housed one. It matters more than the level $\bar{\pi}_r$ because it is a *gradient*: it is what a voucher buys beyond shelter. As it falls (i.e., SUD recovery while unhoused is lower) the gradient steepens, so housing buys more recovery: the private value of a voucher rises and the externality falls. Near one, assistance is a transfer and the moral hazard stands unopposed. Its value is chosen so the model generates the increase in recovery housing produced in Milby et al. (2005), a randomised trial on cocaine-dependent homeless adults: an average of 6.60 weeks abstinent in the housed arm against 4.48 in the homeless, so housing raises recovery by 47.3 percent.

g**, the treatment effect on recovery.** This is the instrument through which Treatment First reaches households that already carry a SUD, which is what matters for short-run impacts, since on implementation some households are already in the SUD state. Raise it and treatment is more likely to cure a SUD, so a household that carries the disorder is more likely to be cured, and therefore the

private cost falls, with the household choosing to go to treatment spending less of its life carrying one. The externality falls with it, because a child who acquires a disorder is also more able to address their disorder, so what its parent hands on costs it less of its own life. It is set externally at 1.35, the odds ratio in Milby et al. (2005).

$\bar{\tau}^u$**, the utility cost of treatment.** Take up is a choice, not a rule, and this is what makes households decline treatment they might gain from in the future. When it rises, fewer opt for treatment, so the externality rises, since more people remain with a SUD and thus more children are born with lower incomes, which leads to higher SUD incidence, and the private cost rises too, since treatment is costly. It is calibrated internally to treatment take-up: 24.5 percent of adults with a SUD received substance use treatment in the past year in NSDUH 2024.

Externalities and distortions. The parameters in this block decide the strength of the externalities and distortions. This affects the ranking between Housing First and Treatment First, since the private comparison already favours Housing First, holding externalities and distortions aside. The first three are what a parent's disorder and a parent's homelessness cost the child, and the fourth is what the homelessness of strangers costs everybody. All four are externalities in the strict sense: one household's choice enters another household's utility directly without being internalised by the household making the decision, and it would still do so even if the programmes were paid for with a non-distortionary tax, such as a lump sum tax. The last is not an externality but a distortion, and it closes the block for that reason: it says what it costs to raise the revenue either programme needs rather than whose utility a choice lands in.

ω_c**, the parent's weight on its child.** A parent bears ω_c of what its own state costs its child, and ω_c is small. As it rises the parent counts more of the child's loss, so the externality left for a policy to correct is smaller. Identifying it in the model requires care, because *every* household here has a child, born after it dies, so on its face it would seem like there is no parent against non-parent comparison to run in the benchmark. I solve this with a counterfactual moment instead. I choose its value so that removing the child's penalty (fixing the other parameters' values) raises drug use by 18.3 percent, the gap between parents and non-parents in NSDUH 2024 read the way the model reads it.

δ_d**, what the child loses to a parent's SUD.** The child does not inherit the disorder, only a worse productivity draw, and that lower expected draw is what makes it likelier to be poor, to use, to acquire a disorder and mental illness consequently, to be unhoused and to die earlier. Increase it and the externality rises, and it loads whichever policy prevents more disorder: Treatment First by curing the parent, Housing First because recovery is easier once housed, while the parent's own private cost is largely unchanged, which is exactly what makes it an externality. It is set externally at 0.24, the earnings loss Nilsson (2017) measures for children exposed in utero to a Swedish alcohol experiment.

δ_h**, what the child loses to homelessness.** It works through the same compounding draw as δ_d, and as it rises it loads whichever policy houses more people: Housing First directly, Treatment First by removing the disorder that leads to homelessness in the first place, again leaving the private cost mostly unchanged. It is set externally at 0.31, the earnings loss from poverty that Chetty et al. (2016) measure: the adult earnings gain from a move to a lower-poverty neighbourhood before age thirteen.

κ_x**, the disamenity of homelessness.** This is the only parameter through which one household's housing status is connected to the utility of another who is not its own child, which is what makes the homelessness rate a genuine fixed-point problem rather than accounting. As it rises the externality from the aggregate headcount grows, which loads whichever policy houses more people. Setting it high should mostly favour Housing First, so I take the upper end estimate for this disamenity as the baseline, and a ranking that still favours Treatment First survives the assumption least kind to it. It is calibrated externally at 0.2 percent of average annual earnings, from the contingent valuation in Loubière et al. (2020).

σ_ℓ**, the curvature on leisure.** The curvature that turns ω_ℓ into an elasticity: together they decide how far hours actually move when the after-tax wage does. The two are not the same dial. ω_ℓ sets what an hour of leisure is worth against consumption, so it fixes the level of hours; σ_ℓ sets how fast that worth falls as leisure is given up, so it fixes how far hours move when the wage changes. One pins the average, the other the response to a tax. At a high curvature hours barely respond and the tax that

funds either policy is close to a lump sum, at a low one raising the same revenues with it costs far more in forgone work. Raise it and labour supply is less elastic, so the distortion falls and both policies look cheaper, without touching the private value or the externality at all. It is set externally to produce a Frisch elasticity of 0.5, from Chetty (2012).

Drug market. One parameter of the drug market belongs here. It is what makes the drug price endogenous, so drug use affects drug demand and the drug price in equilibrium. This is a general-equilibrium offset that runs against Treatment First in particular. Curing households takes them out of the drug market, demand falls and so does the price, and a cheaper drug draws other households in. Some of what the policy removes is handed back through the price, and how much depends entirely on this parameter.

ν_2, the drug supply curvature. I set it so that the price elasticity of supply is 1.46, which is what Mace et al. (2020) estimate for cannabis on Washington State administrative data. To the best of my knowledge, there is no direct estimate for an illicit drug, and cannabis is the closest market there is: it remains a Schedule I substance federally, so even a state licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. When it rises, supply is more elastic, so the price moves less when demand does, and the general-equilibrium offset to treatment shrinks: at a very elastic supply, curing households barely lowers the price, so general-equilibrium effects play little role.

5.2 Standard parameters

The remaining twenty six parameters are more conventional, and I record the values I use so that the calibration can be reproduced. Eight of them I calibrate inside the model, against a moment, because no source reads them off directly; the other eighteen I set from outside, from an estimate or from statute. Table 6 and the entries below run in the same seven blocks, and each entry gives the moment or the source that fixes the parameter together with the value that moment takes in the data. The Model column reports what the model produces against that value, and it does so only for the eight I calibrate internally, since the other eighteen have no target to reproduce.

These parameters are not inert in the experiments. For example, the discount factor governs how heavily a household weighs a death later in life, the dispersion of permanent income determines how many households are poor enough to be at risk of homelessness and of acquiring a disorder, and risk aversion governs what insurance is worth. What separates them from the parameters above is not that they do nothing, but that they are not specific to this paper.

Preferences. The four parameters in this block say what a household wants, and what it is willing to give up to get it.

σ, risk aversion. I set it to the standard value of two, following Attanasio and Weber (2010). Increase it and households want smoother consumption and insurance is worth more, so the voucher is worth more; it also sets the marginal utility of consumption, so it enters every welfare comparison in the paper.

γ, the weight on housing services. This is the Cobb-Douglas share of the bundle that goes on housing, so it fixes what a dwelling is worth against consumption and against the drug. I choose it so that housing takes 26.8 percent of expenditure net of drugs, as shelter plus utilities do in the 2024 Consumer Expenditure Survey. As γ rises housing is worth more against consumption and the drug, households buy more of it, and that expenditure share rises with it.

β, the discount factor. It is chosen so that wealth over output matches the national accounts, a ratio of three, from the Bureau of Economic Analysis (BEA) fixed asset tables against National Income and Product Accounts (NIPA) output, both read at the 2023 vintage, the last year the fixed asset tables cover. As it rises households are more patient, save more, and the wealth to output ratio rises with it.

ω_ℓ, the weight on leisure. It is set so that the model generates average hours worked of 40 percent, from Guvenen et al. (2023a). As it rises leisure is worth more, households work less, and average hours fall

Table 6: Standard parameters

Parameter	Description	Set	Moment and source	Data	Model	Value
Preferences						
σ	Risk aversion	Ext.	Attanasio and Weber (2010)	2.00	n.a.	2.00
γ	Weight on housing services	Int.	Housing over expenditure net of drugs, CEX 2024	26.8%	—	—
β	Discount factor	Int.	Wealth over output, BEA fixed assets	3.00	—	—
ω_ℓ	Weight on leisure	Int.	Average hours worked, Guvenen et al. (2023a)	0.40	—	—
Rental ladder						
h_2	Smallest rental size	Int.	Employment rate of the homeless, HUD 2023	13%	—	—
ς	Curvature, and so h_3	Ext.	Set; the two ends carry the distribution	1.80	n.a.	1.80
h_4	Largest rental size	Int.	Share in largest unit size, ACS 2023	62.7%	—	—
Earnings and employment						
$\varepsilon(j)$	Age profile of efficiency	Ext.	Hump peaking at age 56, Greenwood et al. (2025)	profile	n.a.	profile
σ_y	Dispersion of productivity	Int.	Bottom half's share of earnings, SCF 2022	17.2%	—	—
$\bar{\pi}^{eu}$	Job separation rate	Int.	Unemployment rate, BLS CPS 2014–24	4.85%	—	—
$\bar{\pi}^{ue}$	Job finding rate, annual	Ext.	Krueger et al. (2014)	0.55	n.a.	0.55
Survival and mental illness						
$\bar{\phi}(j)$	Baseline survival by age	Ext.	SSA period life table, ages 26 to 85	profile	n.a.	profile
$s_m(j)$	Age profile of SMI prevalence	Ext.	Share with SMI, NSDUH 2024	5.6%	n.a.	5.6%
Interest, taxes and public spending						
r	Risk free rate	Ext.	Kaplan et al. (2020)	0.03	n.a.	0.03
ξ	Tax progressivity	Ext.	Heathcote et al. (2017)	0.85	n.a.	0.85
τ_k	Capital income tax	Ext.	US long run rate on gains and dividends	0.15	n.a.	0.15
τ_ℓ	Tax level	Int.	Average labour income tax, McDaniel (2007)	22.4%	—	—
Social security benefits and programmes' rules						
ψ_R	Retirement replacement rate	Ext.	SSA, share of the productivity state	0.40	n.a.	0.40
ψ_{ui}	UI replacement rate	Ext.	DOL statutory replacement rates	0.50	n.a.	0.50
$\bar{y}$	Cap on the earnings base	Ext.	DOL state benefit maxima	1.00	n.a.	1.00
κ_V	Voucher means test	Ext.	HUD rule	0.50	n.a.	0.50
κ_r	Tenant's share of the rent	Ext.	HUD rule	0.30	n.a.	0.30
ϱ	Food stamp taper rate	Ext.	SNAP statute, 7 CFR 273.10	0.30	n.a.	0.30
Drug market						
J_E	Last experimentation age	Ext.	Model age 55, that is age 80	55	n.a.	55
ν_1	Drug supply shifter	Ext.	Normalisation, so $p_v = 1$	norm.	n.a.	norm.
p_v	Price of a unit of drugs	Ext.	Clears the drug market in the benchmark	1.00	n.a.	1.00

Note: n.a. in the Model column marks a parameter that is set from outside rather than calibrated inside the model. For those the model produces no internal target to compare with the data, so the cell reads n.a. rather than a dash, which keeps them distinct from the internally calibrated parameters.

with it.

Rental ladder. These three parameters build the grid of dwelling sizes a renter chooses from, and they fix how the rungs are spaced. Each rung answers to one moment.

h_2**, the smallest rental size.** The size of the cheapest dwelling is what decides how many households take the bottom rung rather than move up, so its value is chosen so the model generates the 13 percent of housed households living in the smallest unit size in the 2023 ACS, Table B25042. Raise it and the bottom rung is larger and dearer relative to the rung above, which is nearly the same dwelling at nearly the same price, so more households stay on it and that share rises.

ς**, the curvature of the rental ladder, which determines** h_3**.** The curvature places the middle rung, a fraction $(1/2)^\varsigma$ of the way from h_2 to h_4. It is set rather than calibrated: the three shares sum to one, so matching the smallest with h_2 and the largest with h_4 matches the middle as well, and a third instrument on it would have no moment of its own left to answer to.

h_4**, the largest rental size.** The top of the ladder, calibrated on the share of the housed in the largest unit size, 62.7 percent in the same table B25042. Three shares carry two free numbers, so with ς holding the middle share this one pins the whole distribution and the smallest category follows. The universe is occupied units, so the model counterpart conditions on being housed, and the homeless are in neither denominator.

Earnings and employment. These five parameters determine what a household is expected to earn over its life, how much of that is permanent, and how it moves between work and unemployment.

$\varepsilon(j)$**, the age profile of efficiency.** The deterministic hump in earnings over the life cycle, taken as the exponential of a quadratic in age used by Greenwood et al. (2025), $\varepsilon(j) = \exp\{-(j-1)^2/1800 + (j-1)/30\}$ with $j = 1$ at age 26. It peaks at age 56, having risen 65 percent from age 26, and then falls 4 percent from the peak to age 65, the last working age. Without it every household would earn the same at every age, and the ages at which households are poor, and so at risk of homelessness and of using, would be wrong at both ends of working life.

σ_y**, the dispersion of productivity.** Productivity is permanent: a household draws it once and keeps it for life, exactly as in the stylized model, so all that is needed is the spread of that draw. It is calibrated, to the share of earnings held by the bottom half of the income distribution among working households. It is what decides how many households are poor enough to be at risk of homelessness and of acquiring a SUD. In the 2022 SCF, households aged 26 to 65 with positive wage and salary income in the poorer half of the distribution hold 17.2 percent of total earnings, and σ_y is chosen to reproduce that share. The share falls in σ_y.

$\bar{\pi}^{eu}$**, the job separation rate.** Chosen to reproduce the aggregate unemployment rate in the BLS Current Population Survey (CPS) over 2014 to 2024, 4.85 percent, given the job finding rate set externally below. Raise it and more households lose work each year, so the unemployment rate rises with it.

$\bar{\pi}^{ue}$**, the annual job finding rate.** The probability that an unemployed household finds work by the end of a period, which I set to 0.55 from the annual exit rate Krueger et al. (2014) estimate on matched CPS panels.

Survival and mental illness. These two parameters are age profiles that I take from outside the model rather than generate inside it.

$\bar{\phi}(j)$**, baseline survival by age.** The most recent SSA period life table, Table 4.C6, for ages 26 to 85, with the sexes combined by survivor weight rather than averaged. This is the profile every mortality multiplier in Section 5 multiplies, so it is the base the SUD, homelessness and mental illness hazards are measured against.

$s_m(j)$**, the age profile of SMI prevalence.** The target is the share of adults carrying SMI at each age j in NSDUH 2024, which integrates to 5.6 percent across all ages; that is the overall level and the profile is what is matched, age by age. It sits here rather than in Section 5 because it is the baseline profile, in the same way the survival table is: what this paper adds is ι, the multiplier a SUD puts on that onset, and that is where the interaction lives.

Interest, taxes and public spending.

r**, the risk-free rate.** I set it to three percent, from Kaplan et al. (2020).

ξ**, tax progressivity.** The exponent on income in the tax schedule of Section 4.3 is set to 0.85 from Heathcote et al. (2017). It determines the progressivity of the labour income tax. That in turn decides how much a programme's bill falls on the households it is meant to help, and how much on everybody else.

τ_k**, the capital income tax.** I set it to fifteen percent, consistent with the statutory United States rate on long-term capital gains and qualified dividends for the middle of the income distribution, which the Internal Revenue Service (IRS) sets out in Topic 409 (`https://www.irs.gov/taxtopics/tc409`).

τ_ℓ**, the level of the labour income tax.** It is calibrated so that the average labour income tax rate the model produces is 22.4 percent, which is what McDaniel (2007) measures for the United States and what Guvenen et al. (2023a) use. It then moves with the government budget in every counterfactual, which makes both programmes distortionary.

Social security benefits and programmes' rules.

ψ_R**, the social security replacement rate.** Retirement income is $\psi_R = 0.40$ times the base productivity state for the last $J - J_R$ years of life, from the SSA. As it rises the old are richer (at the expense of the young, who are subject to higher distortionary taxes), so the value of a life still to be lived rises and with it the cost of a disorder that shortens it.

ψ_{ui} **and** $\bar{y}$**, unemployment insurance.** These two set the UI replacement rate and the cap on the earnings it is paid on. I set them to fifty percent and one times average income, respectively, taking both from the statutory replacement rates and state benefit maxima that the Department of Labor compiles in its comparison of state UI laws (`https://oui.doleta.gov/unemploy/statelaws.asp`).

κ_V **and** κ_r**, the voucher rules.** These two set the income cutoff for eligibility and the tenant's contribution, and I take both from HUD's own rules. I set them to half of median income and thirty percent, respectively. The first is the very low-income limit HUD publishes annually with its income limits (`https://www.huduser.gov/portal/datasets/il.html`); the second is the share of rent a tenant pays when a voucher covers the rest, under the Housing Choice Voucher programme (`https://www.hud.gov/helping-americans/housing-choice-vouchers`).

ϱ**, the food stamp clawback rate.** This is the rate at which food assistance is withdrawn as income rises. I set it to thirty percent, which is the statutory rule: a household's monthly allotment is the maximum allotment for its size reduced by thirty percent of its net monthly income (7 CFR 273.10). It matters beyond the level of the transfer, because it is the implicit marginal tax the programme puts on earnings for the households the paper is about: an extra dollar earned costs a household on food assistance ϱ of a dollar of its allotment.

Drug market. These three parameters describe the rest of the drug market, closing the loop on the elasticity parameter explained in Section 5: the first fixes who can enter it, and the other two fix the units in which its price and quantity are measured.

J_E**, the maximum age at which use can begin.** This restricts only households that do not have a SUD, not ones that already do. I set it to fifty five in model terms, which corresponds to age 80 in the data. It is there so that use cannot begin in the last few years of life, where almost no continuation value remains to be lost and the private cost of using is therefore close to zero. That is an artefact of the horizon rather than a feature of the world: to keep the age dimension manageable the model imposes certain mortality after age 85, so a household near the end knows death is coming with certainty and a disorder costs it almost nothing. Without the cap the model would let use begin exactly there, and at the final age it would be worse than that, since a household that acquires the disorder in the last period dies at the end of it whatever the hazard does: the disorder carries literally no consequence and only the flow benefit of using, which cannot be a reasonable description of the choice either.

ν_1**, the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark, and held fixed across experiments. It normalises units, not behaviour.

p_v**, the drug price.** Set to one by normalisation in the benchmark, but free in every counterfactual. It is an equilibrium object rather than a dial.

5.3 Validation

None of the moments below is fitted, with one qualification: the homelessness rate's overall level is a calibration target, so what is tested in the first row is the profile across the five age bands rather than the level of it. Each is produced endogenously by a model with sixty single ages, and each is a place the model can be caught. Section 9 of `data/DATA_APPENDIX.md` in the replication package derives the data side of each, and it is the place to go for everything needed to replicate them: the dataset, the year, the universe, the computation, and the three cases where the universe rather than the arithmetic decides the number. The package ships those datasets as well, so none of them has to be downloaded first. Two shares of the earnings distribution come out alongside the one σ_y is chosen on and are untargeted: the model's top tenth holds 38.5 percent of earnings against 41.1 in the data, and its bottom tenth 1.1 percent against 0.99. The last three rows are the intergenerational transmission the paper turns on, and the homelessness one needs a word about its construction. Bassuk et al. (1997) is a case-control study of 220 homeless mothers and 216 low-income housed mothers, so both arms are poor and the comparison isolates transmission from the poverty that accompanies it. The share whose primary female caretaker had ever been homeless was 14.2 against 8.3 percent, and for the male caretaker 10.3 against 4.4, which are odds ratios of 1.8 and 2.4. A ratio taken against the general population instead would be far larger and would credit the housing penalty with

Table 7: Moments the model is not asked to match

Moment	Data	Source
Homelessness rate, 25–34, 35–44, 45–54, 55–64, 65+	0.316, 0.338, 0.291, 0.250, 0.077 percent	HUD PiT 2024
Shares of the homeless across those bands	26.0, 27.2, 21.0, 18.4, 7.5 percent	HUD PiT 2024
Returns to homelessness within 24 months	17.4 percent	HUD SPM 2024
Drug poisonings as a share of deaths, 25–44	about 20 percent	CDC WONDER
Earnings shares, bottom tenth and top tenth	0.99 and 41.1 percent	SCF 2022
Bottom half's share of net worth	2.52 percent	Fed DFA
Housing costs above 30 and 50 percent of income	40.6 and 17.5 percent	SCF 2022
Tenant contribution rate	31.2 percent	HUD PSH 2023
Voucher holders with wages as main income	28 percent	HUD PSH 2023
Income with a moderate or severe SUD	71.1 percent of average	NSDUH 2024
Income of drug users	74.0 percent of average	NSDUH 2024
Labour force participation, with a SUD against without	65.8 against 61.5 percent	NSDUH 2024
Homeless at treatment admission	14.5 percent	TEDS-A 2022
Psychiatric problem alongside SUD at admission	43.1 percent	TEDS-A 2022
Working at all, homeless in treatment	6.6 percent	TEDS-A 2022
Price elasticity of demand, participation	-0.63	Saffer and Chaloupka (1999)
Price elasticity of demand, intensive margin	-0.55	Midpoint, Section 2
Odds a homeless adult had a homeless caretaker	1.8 to 2.4	Bassuk et al. (1997)
Odds a child of a parent with a SUD acquires one	1.44	Mellentin et al. (2016)
The same, United States	2.51	Yoon et al. (2013)

everything else that differs between poor children, which Cobb-Clark and Zhu (2015) show is substantial: among the households of the Journeys Home panel (Scutella et al., 2012), every one of them on income support, those first homeless as children were more than twice as likely to have been in foster care.

Every parameter is disciplined, either by a moment the model is asked to reproduce or by an estimate taken from outside it, and Tables 5 and 6 name which for each. Three rest on a single study rather than on a literature: the treatment effect on recovery g, the recovery multiplier for SMI μ_m and the drug supply curvature ν_2. Of these only g carries a headline result, so it is reported as a range rather than as a number.

6 The welfare criterion

Every welfare number in this paper is a CEV, and the convention matters enough to state exactly. Write V^b for lifetime value under the benchmark and V^c for lifetime value under a counterfactual policy. The CEV Δ is the proportional change in consumption, applied at every date and in every state, that makes a newborn under the benchmark as well off as the same newborn under the counterfactual:

$$V^b\big((1+\Delta)\,c\big) \;=\; V^c(c). \tag{40}$$

A positive Δ therefore means the counterfactual is worth that much of a permanent consumption gain, and a negative one that it costs that much.

Whose welfare. The household considered is a newborn, and the value is taken before any of its draws are realised, so Δ is an expectation over the productivity draw and everything that follows from it. The expectation is taken under each policy's own equilibrium rather than a common one, since the penalties that set a child's starting productivity depend on what happened to its parent, so the distribution a newborn is drawn from is itself something the policy moves. This is the standard criterion in this literature (Conesa et al., 2009; Guvenen et al., 2023b).

Two horizons are used, and they are the same object measured at different dates: a newborn on the date the policy is implemented, who lives through the transition, and a newborn in the new steady state. Because both are the ex-ante value of a household at age one, any difference between them is the transition itself rather than a change in who is being counted.

The stylized model has a closed form and the quantitative model does not. With log utility over consumption and housing, scaling consumption by $1+\Delta$ at every date adds $K\log(1+\Delta)$ to lifetime

value, where K collects the weight on consumption and the discounted periods a household expects to live, $K = (1 - \gamma - \theta)(1 + \beta\bar{\phi})$. Equation (40) then inverts directly to $\Delta = \exp\{(V^c - V^b)/K\} - 1$, which is how every CEV in Section 3.6 is computed.

No such inversion exists in the quantitative model, and three features of flow utility in equation (14) are why. The flow value of being alive $\aleph$ is additive, so it does not scale with consumption. The drug enters through $\log(b + v)$ with a Stone-Geary shifter, so scaling consumption does not scale the drug term proportionately. And food assistance pays exactly the households the paper is about, and it is withdrawn at the rate ϱ rather than one for one, so scaling their consumption scales their resources by less than the scaling factor. Δ is therefore solved numerically, by bisection on the newborn's expected value.

7 Experiments

Plan for this section, not its content. Four steps, set out in full in `code/EXPERIMENT_PLAN.md` and summarised here so the paper's structure is legible:

1. *The optimal generosity of each policy, separately*, over eligibility, the share of rent covered, and the probability of receipt, the last varied continuously so the question of whether the corner is optimal can be answered rather than assumed.
2. *Each optimum against the other policy at the same generosity*, which is what isolates conditionality from generosity. Both optima, because the optimal Housing First design is not the optimal Treatment First design. Reported under both readings of "the same": same design, and same cost.
3. *Transition against the long run*, for whichever policy wins, because a trial that runs six to twenty four months sees the first periods of a transition and not a steady state.
4. *The channel shutdown ladder at the optimum*, in both horizons: the child externality, the aggregate externality, the labour supply response that carries the tax distortion, the drug price response, and the cost of administering the test. The rungs do not commute, so a fixed order is reported together with the interaction term.

The parameters the experiment introduces. Three of the model's parameters do nothing in the benchmark and are therefore set here rather than in Section 5: nothing is screened when no voucher is conditional, so none of them binds until Treatment First is imposed. The detection rate is $p_{\text{det}} = 0.9$ and the false positive rate $p_{\text{fp}} = 0.05$, both set rather than measured, since no programme publishes a detection rate for a drug screen in a housing programme; the counterfactuals report the range around them rather than resting on the point. The government pays $c_{\text{test}} = 0.031$ percent of average earnings for every household it screens. That last figure is a lower bound on the administrative burden and is meant as one: it is the assay cost in the state fiscal notes of the public assistance programmes that have tried testing, and the programme level figures are far larger once staffing is counted, so the cost of administering the test is a rung of the shutdown ladder rather than a settled number.

8 Discussion

Plan for this section, not its content. Four things to say once the numbers exist:

- *What the trials can and cannot settle.* If the ladder shows the externality nearly absent at implementation and growing with the horizon while the tax distortion bites immediately, that is the reconciliation: the trials are well powered for the margin that moves first and silent on the one that decides the ranking.
- *Which channel carries the result.* The paper insists on separating an externality from the distortions, and the ladder is what makes that separation a number rather than a claim. If the aggregate disamenity turns out to be small next to the intergenerational one, saying so is part of the point.

- *What would overturn it.* The three undisciplined parameters above, and the value of ω_c, are where a reader should push. Report what each would have to be for the ranking to flip.
- *What the model does not have.* Neighbourhood effects on the child are priced through δ_h rather than modelled spatially; treatment capacity is not rationed; and the detection rates are set rather than estimated.

9 Conclusion

Plan for this section, not its content. Short. The moral hazard the debate argues about is real and is not an inefficiency, so it cannot carry the case for conditionality on its own. The case has to rest on an externality, on the distortions that fund the programme, or on the programme's own distortions, and those are three different claims. This paper prices the first and finds it intergenerational: what a parent's disorder and homelessness cost the child who replaces it. Close on what that implies for a policy debate currently being settled by executive order.

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